

Relevance-Based Prediction: Factor Selection for Saugus Schools

Four outcomes: MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, Education Budget Share

One RBP run per outcome on ALL candidate variables — no feature pruning (faithful to Kritzman)

Exhibit 5 importance is descriptive: positive = helps, ≈ 0 = benign / diversified away (fn. 12), < 0 = harmful

Saugus analyzed as the prediction task; most/least relevant towns identified per Exhibit 4

Leave-one-out LOO r validates the predictions across all MA districts

Model Summary (all candidates used — no pruning)			
MCAS Grades 3-8	Dropout Rate	MCAS Grade 10 (ELA)	Education Budget Share
Predicted on 19 candidates	Predicted on 19 candidates	Predicted on 19 candidates	Predicted on 19 candidates
Positive importance: 18 of 19	Positive importance: 18 of 19	Positive importance: 18 of 19	Positive importance: 18 of 19
LOO r = +0.902	LOO r = +0.838	LOO r = +0.900	LOO r = +0.801

Pre-Specified Candidate Pool — Feature Reduction

Features selected on domain grounds before running RBP. Reduces K/N ratio from ~0.19 to ~0.09 (target ≈ 0.085).

Kept — 14 pure predictors

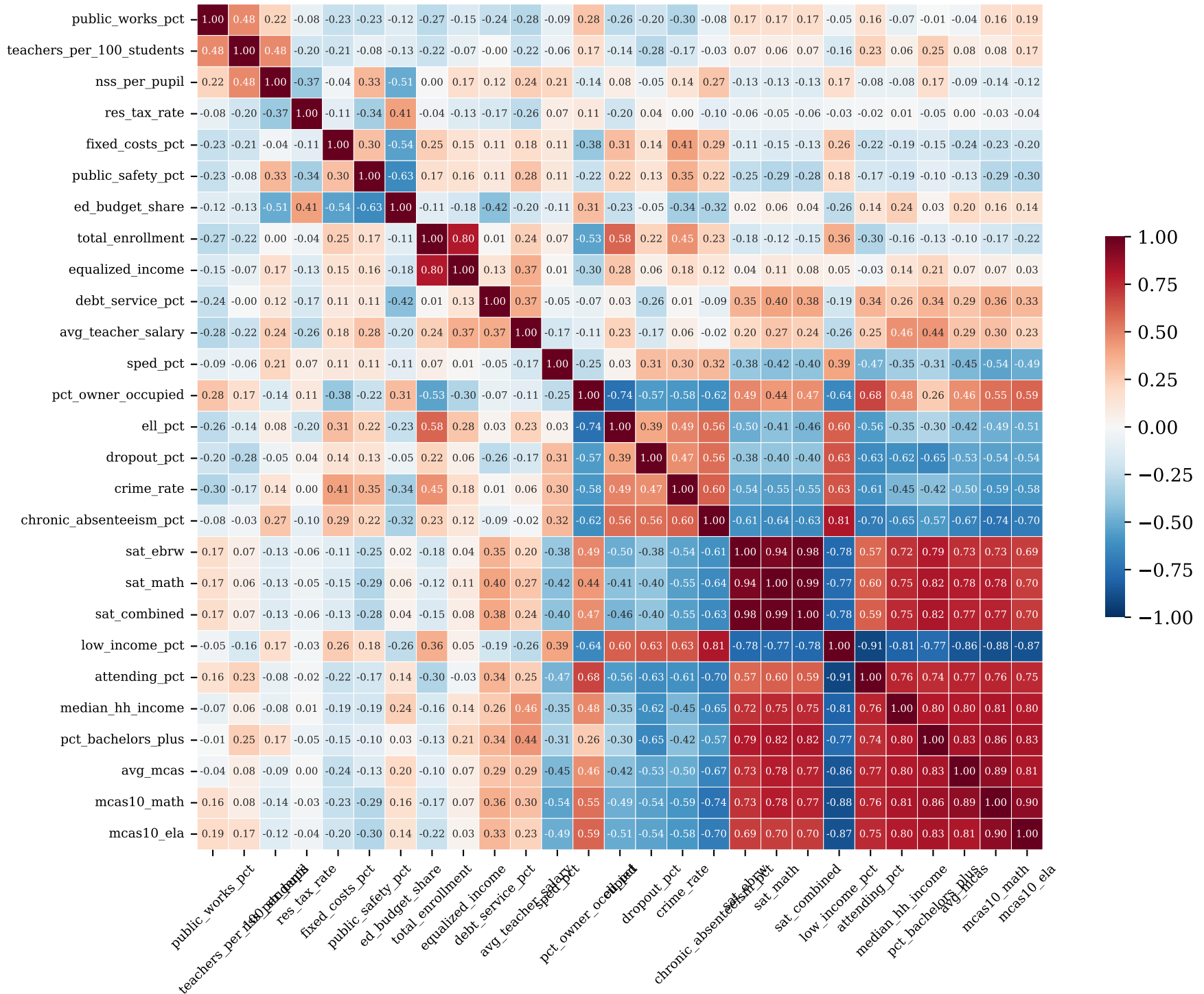
Group	Feature	Measures
Poverty	low_income_pct	% low-income students
Wealth	median_hh_income	Median household income
Wealth	equalized_income	Equalized property value/capita
Human capital	pct_bachelors_plus	% adults with bachelor's+
Housing	pct_owner_occupied	% owner-occupied homes
Community	crime_rate	Crime incidents per capita
Community	res_tax_rate	Residential tax rate
Engagement	chron_absenteeism_pct	% chronically absent
Demographics	ell_pct	% English language learners
Demographics	sped_pct	% special education students
Size	total_enrollment	District enrollment
Staffing	teachers/100_students	Teachers per 100 students
Staffing	avg_teacher_salary	Average teacher salary
Spending	nss_per_pupil	Net school spending/pupil

Dropped — 12 redundant/problematic features

Dropped	Kept instead	Reason
high_needs_pct	low_income_pct	$r=+0.981$ — near-duplicate
acs_poverty_pct	low_income_pct	$r=+0.87$ — redundant poverty
foundation_budget_pp	nss_per_pupil	$r=+0.92$ — redundant spending
ch70_per_pupil	equalized_income	$r\approx-0.87$ — inverse wealth proxy
teacher_fte	total_enrollment	$r=+0.994$ — linear with enroll
total_population	total_enrollment	$r=+0.953$ — size proxy
teachers_per_100_fte	t_per_100_students	$r=+0.872$ — duplicate ratio
teacher_spending_pp	nss_per_pupil	$r\approx+0.85$ — redundant spending
gf_exp_per_capita	res_tax_rate	$r\approx+0.75$ — redundant municipal
pct_65_plus	—	Weak signal, no policy lever
com_tax_rate	res_tax_rate	$r\approx+0.65$ — commercial tax dup
mcas10_math	mcas10_elar	$r=+0.90$ — one grade-10 measure

Correlation threshold for deduplication: $|r| > 0.87$. Cross-outcome features (avg_mcas, mcas10_ela, dropout_pct, attending_pct) handled separately per model via exclusion rules.

Feature Correlation Matrix — All Candidates (hierarchically clustered)
Features ordered by |correlation| — dark red/blue clusters = high redundancy

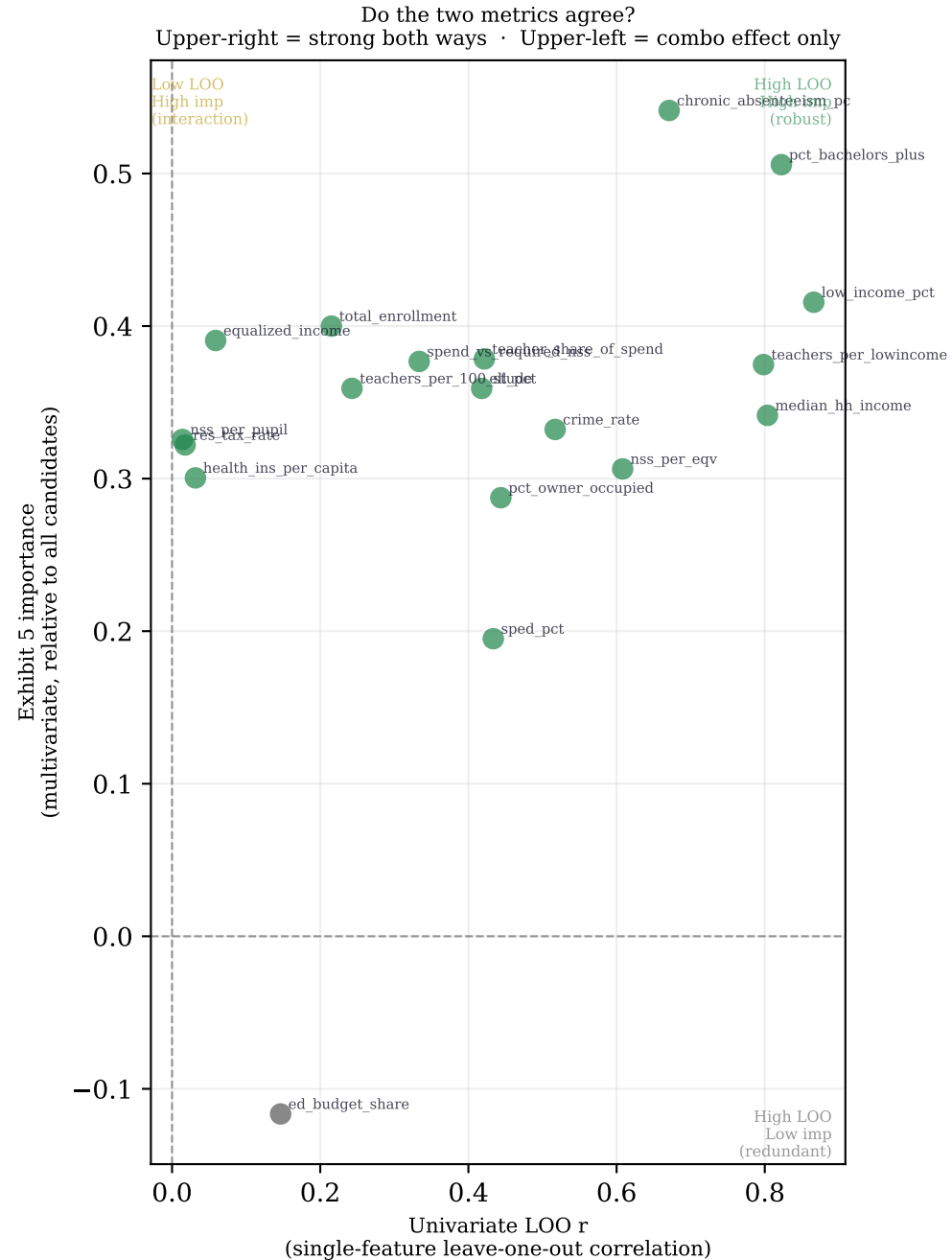
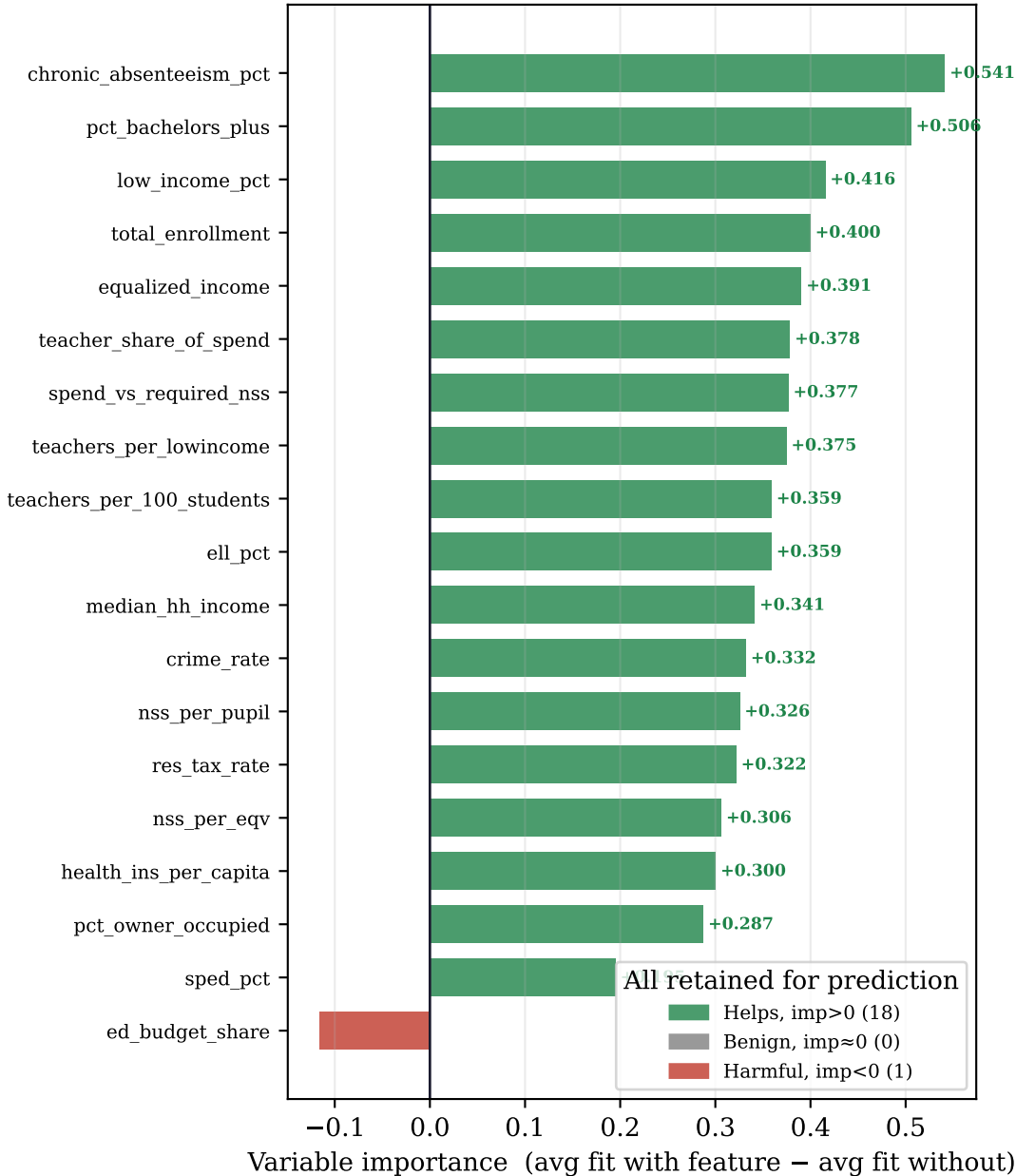


Pearson r. Hierarchical average-linkage clustering on $|1-r|$ distance. Features with $|r| > 0.87$ collapsed to one representative in the candidate pool.

Variable Importance (Exhibit 5): MCAS Grades 3-8

19 candidates, all used for prediction · n_random=3000 · top-3 NOT seed-stable · 18 help (imp>0) · 0 benign (≈0) · 1 harmful (imp<0)

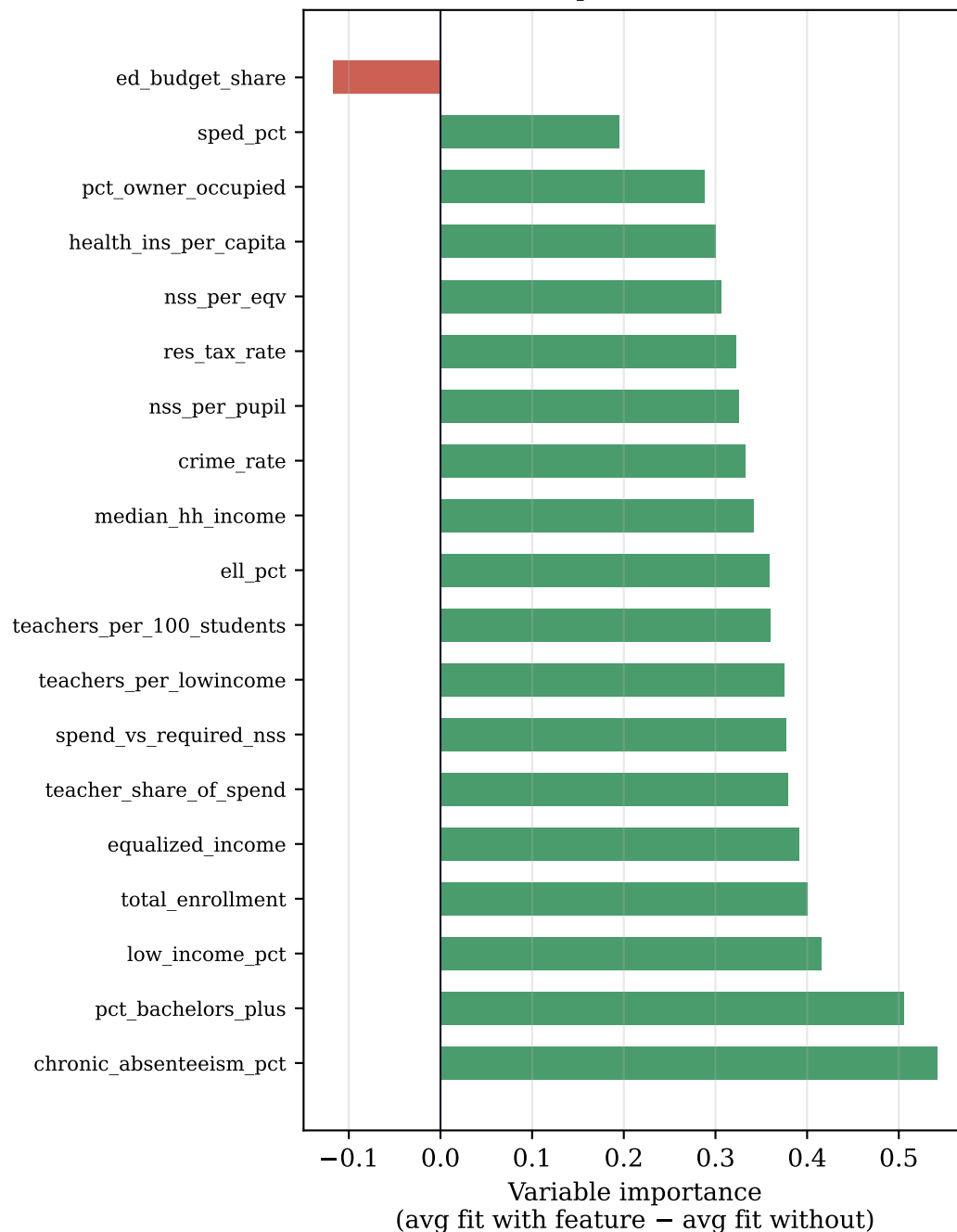
All Candidates — Variable Importance (all retained)
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful



Saugus RBP Analysis: MCAS Grades 3-8

Prediction task = Saugus · Predicted: 23.6pp Actual: 29.0pp Gap: +5.4pp (Fit: 0.1863)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	avg_mcas
Petersham	0.0577	0.40
Nahant	0.0517	0.51
Winchendon	0.0473	0.21
Everett	0.0470	0.18
Eastham	0.0440	0.55

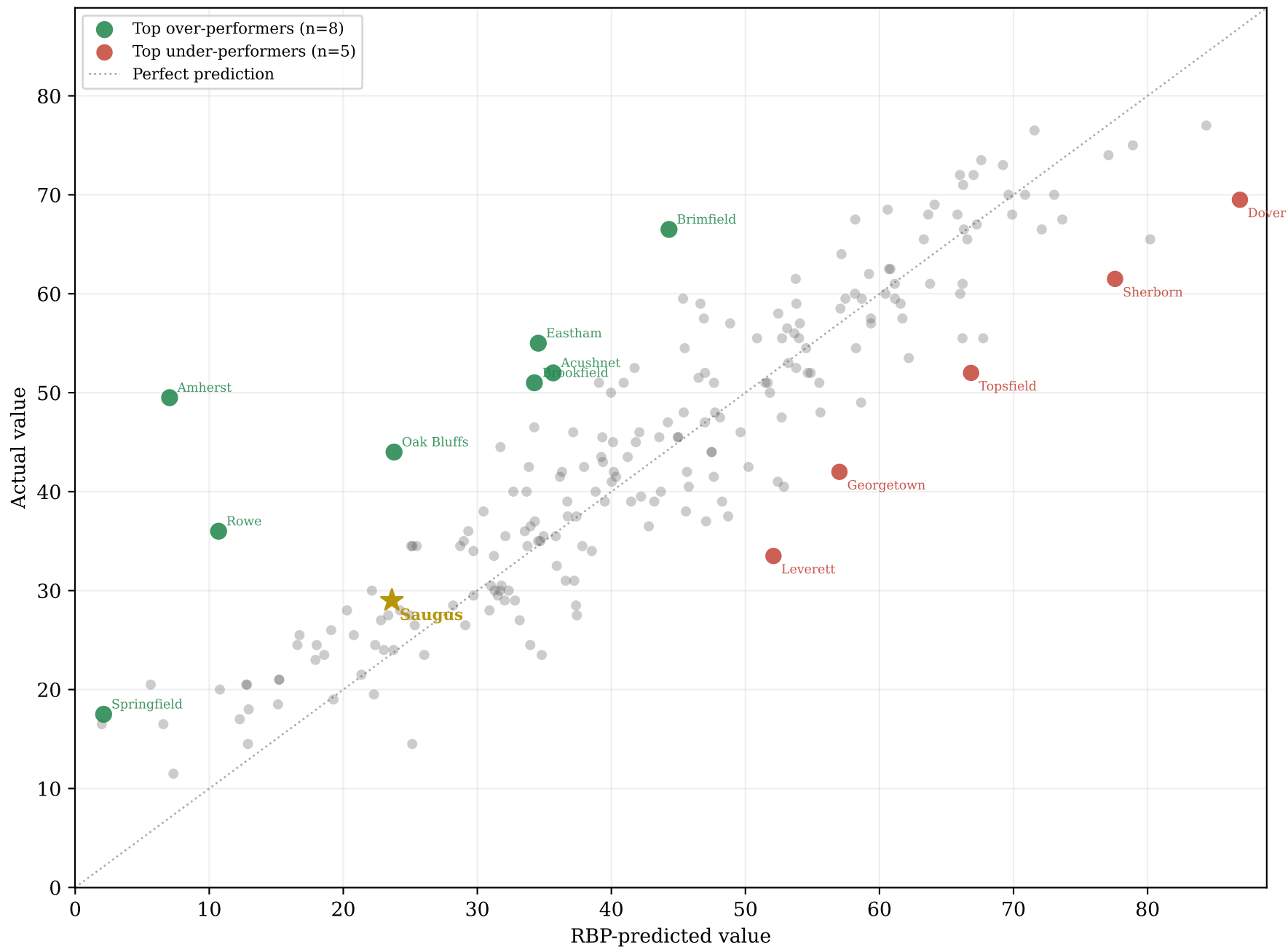
Least Relevant Towns (lowest weight)

Town	Weight	avg_mcas
Ipswich	-0.0129	0.56
Mattapoisett	-0.0108	0.61
Andover	-0.0106	0.66
Mansfield	-0.0106	0.55
Amherst	-0.0103	0.49

Over- and Under-Performers: MCAS Grades 3-8

Saugus gap: +5.4pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grades 3-8 (LOO r = 0.902)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: MCAS Grades 3-8

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Amherst	Rowe	Brimfield	Imp
chronic_absenteeism_pct	31.2	16.7	26.2	20.3	+0.541
pct_bachelors_plus	31.4	65.7	24.5	43.5	+0.506
low_income_pct	46.6	34.4	52.5	31.3	+0.416
total_enrollment	3,084	1,135	86.0	299.0	+0.400
equalized_income	1.17B	826M	15M	151M	+0.391
teacher_share_of_spend	0.35	0.31	0.33	0.46	+0.378
spend_vs_required_nss	1.03	1.76	1.91	1.36	+0.377
teachers_per_lowincome	0.13	0.28	0.20	0.29	+0.375
teachers_per_100_student	5.87	9.80	10.6	9.23	+0.359
ell_pct	13.0	13.6	0.00	1.10	+0.359
median_hh_income	100,819	56,905	87,917	101,477	+0.341
crime_rate	2,813	1,013	0.00	700.1	+0.332
nss_per_pupil	17,369	31,268	34,298	20,046	+0.326
res_tax_rate	10.7	18.5	5.01	14.6	+0.322
nss_per_eqv	0.07	0.35	0.03	0.12	+0.306

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	29.0	23.6	+5.4pp
Amherst	49.5	7.0	+42.5pp
Rowe	36.0	10.7	+25.3pp
Brimfield	66.5	44.3	+22.2pp
Eastham	55.0	34.5	+20.5pp
Oak Bluffs	44.0	23.8	+20.2pp
Brookfield	51.0	34.3	+16.7pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grades 3-8

Predicted 23.6pp · Actual 29.0pp · Gap +5.4pp (higher is better) · Saugus ranks 54 of 219 MA districts on this measure

The Bottom Line

Saugus's MCAS grades 3–8 proficiency (ELA + Math, meeting/exceeding) is 29.0%, 5.4pp better than its demographic prediction of 23.6% — ranking 54 of 219 MA districts, in the top 25%. A district with Saugus's income, poverty, and enrollment profile is predicted to land near this level.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.54
Adults with a bachelor's degree+	31.4%	46.9%	+0.51
% students from low-income families	46.6%	30.4%	+0.42
Total district enrollment	3,084	1,669	+0.40
Equalized property value per capita	\$1.17B	\$935M	+0.39

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	19.1%	-12.1pp
Adults with a bachelor's degree+	31.4%	35.4%	+4.0pp
% students from low-income families	46.6%	36.9%	-9.7pp
Total district enrollment	3,084	386	-2,698
Equalized property value per capita	\$1.17B	\$190M	-981M

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

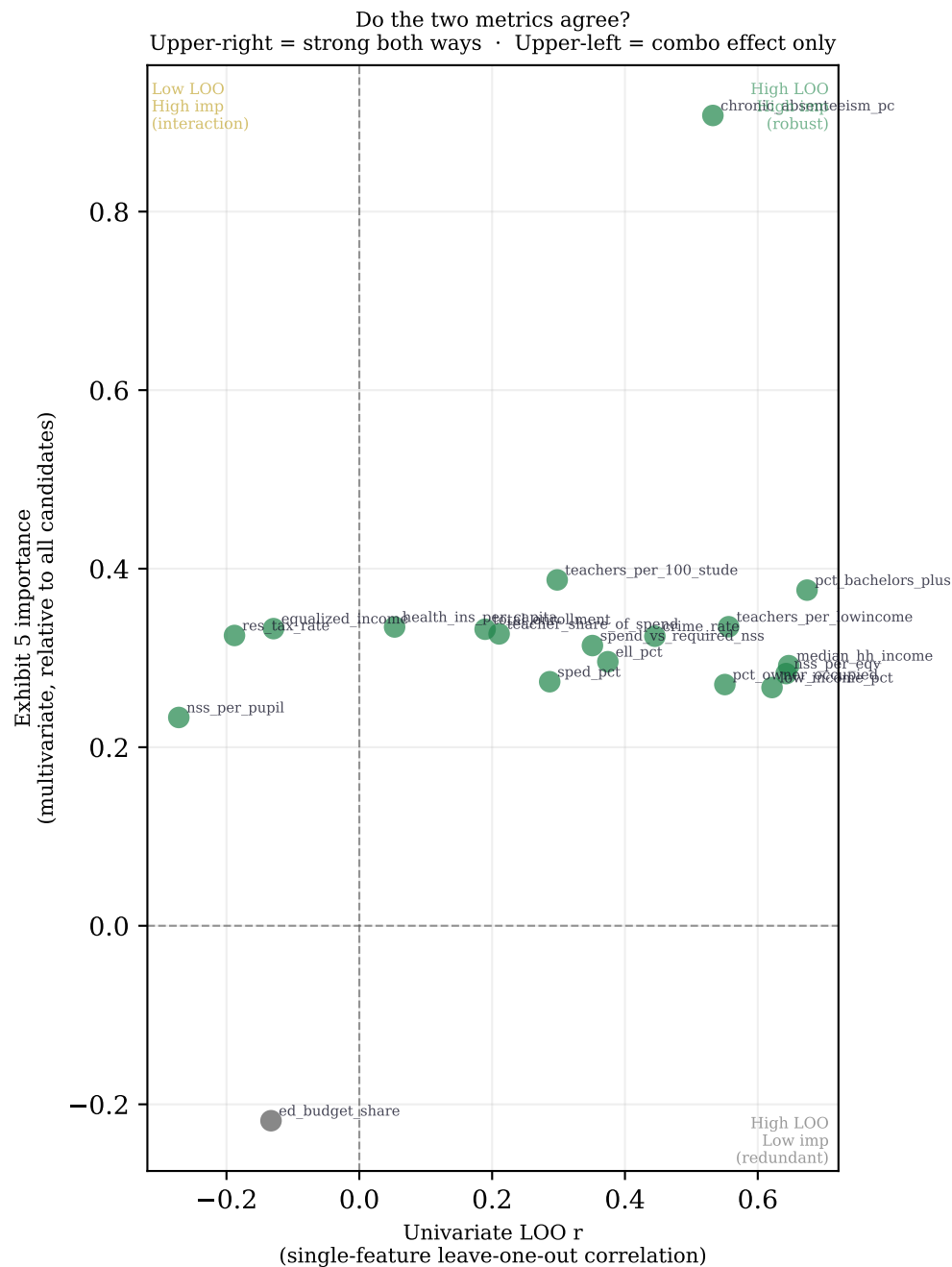
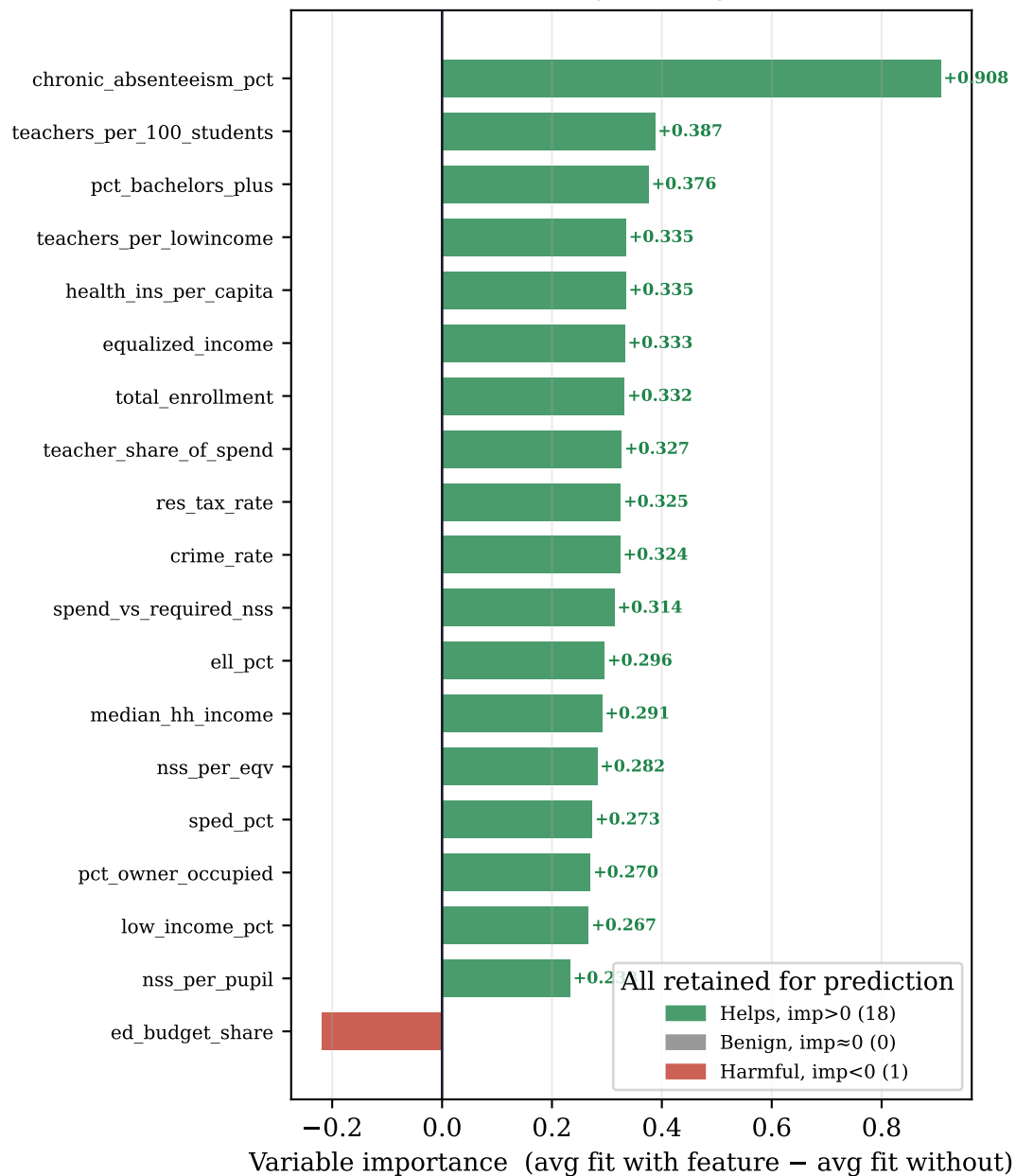
The Takeaway

Saugus does better here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is equalized property value per capita (-981M, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) The drivers below are where Saugus differs most from the towns that beat their own predictions — the most plausible places to move this result.

Variable Importance (Exhibit 5): Dropout Rate

19 candidates, all used for prediction · n_random=3000 · top-3 NOT seed-stable · 18 help (imp>0) · 0 benign (≈0) · 1 harmful (imp<0)

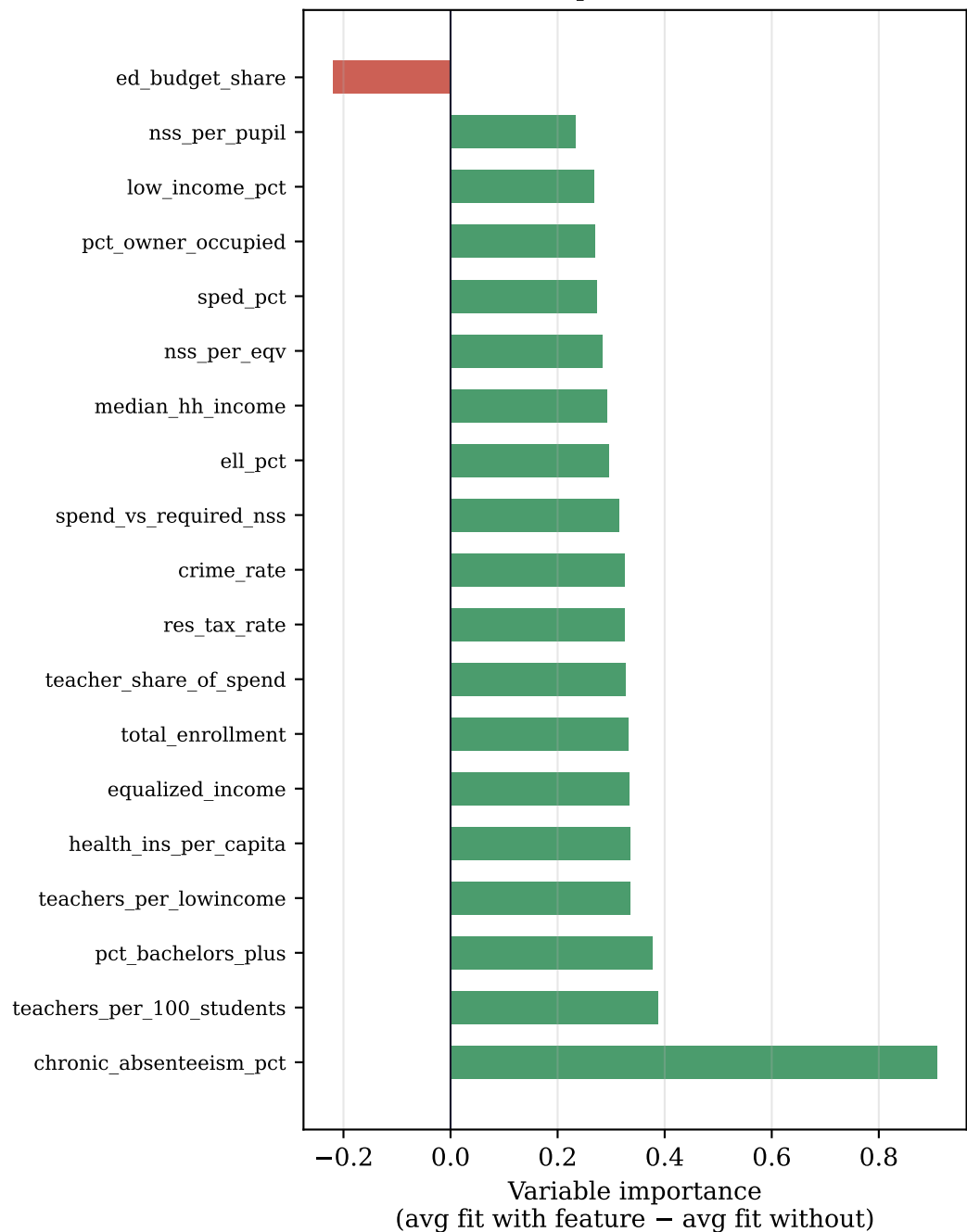
All Candidates — Variable Importance (all retained)
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful



Saugus RBP Analysis: Dropout Rate

Prediction task = Saugus · Predicted: 4.1pp Actual: 2.9pp Gap: -1.2pp (Fit: 0.1539)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	dropout_pct
Everett	0.0943	5.20
Gardner	0.0795	2.90
Winchendon	0.0786	7.50
Gloucester	0.0641	2.50
Dracut	0.0639	1.40

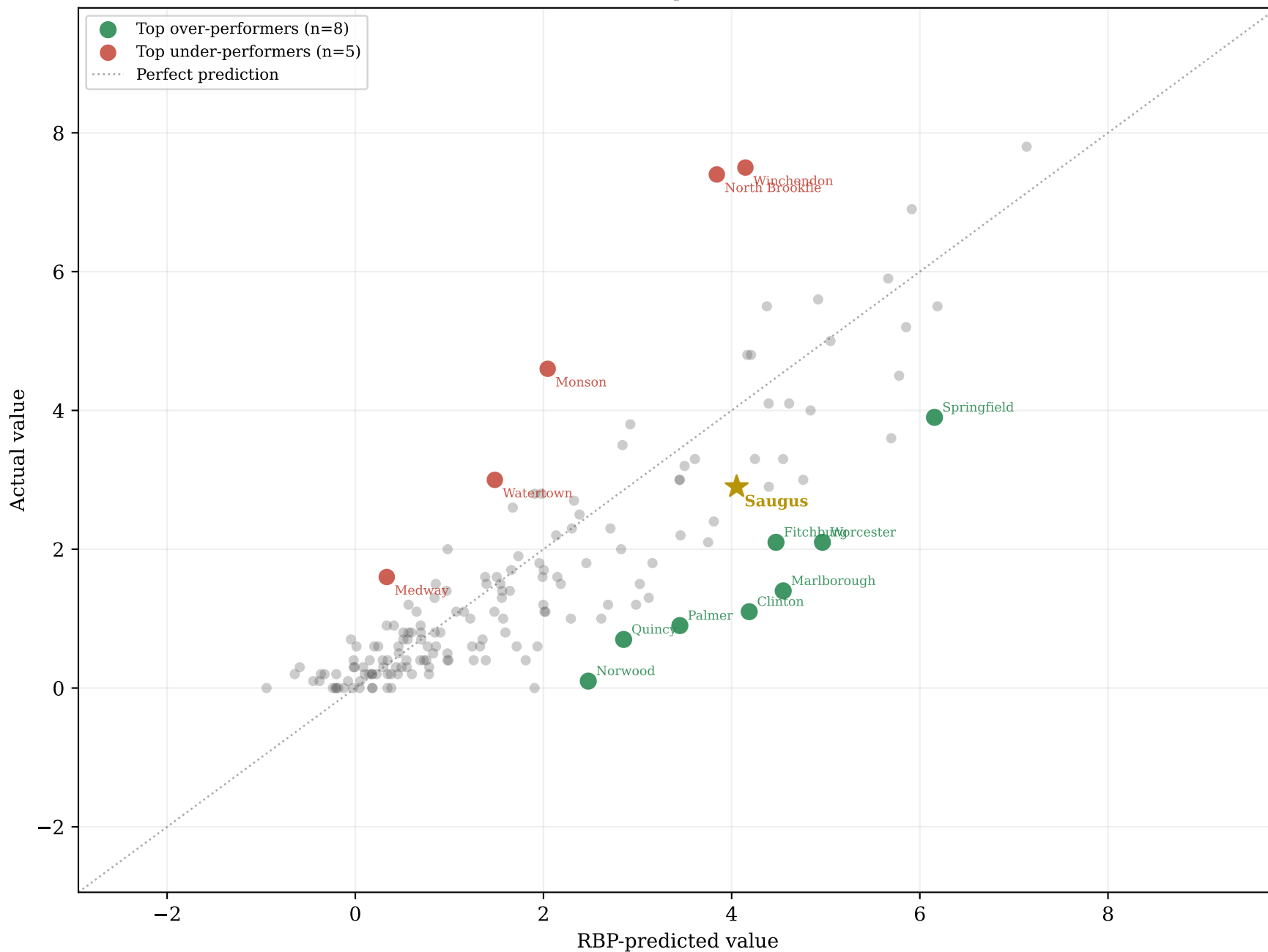
Least Relevant Towns (lowest weight)

Town	Weight	dropout_pct
Andover	-0.0194	0.20
Foxborough	-0.0191	0.70
Newburyport	-0.0188	0.00
Bedford	-0.0184	0.20
Natick	-0.0182	0.70

Over- and Under-Performers: Dropout Rate

Saugus gap: -1.2pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — Dropout Rate (LOO r = 0.838)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: Dropout Rate

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Marlborou	Clinton	Worcester	Imp
chronic_absenteeism_pct	31.2	23.5	7.70	28.3	+0.908
teachers_per_100_student	5.87	7.36	7.31	7.40	+0.387
pct_bachelors_plus	31.4	40.6	35.1	34.1	+0.376
teachers_per_lowincome	0.13	0.13	0.14	0.10	+0.335
health_ins_per_capita	588.1	794.6	362.4	700.1	+0.335
equalized_income	1.17B	1.65B	514M	5.00B	+0.333
total_enrollment	3,084	5,572	2,237	27,606	+0.332
teacher_share_of_spend	0.35	0.37	0.39	0.38	+0.327
res_tax_rate	10.7	10.2	13.1	13.8	+0.325
crime_rate	2,813	3,695	324.5	5,437	+0.324
spend_vs_required_nss	1.03	1.11	1.08	1.10	+0.314
ell_pct	13.0	32.9	19.7	30.4	+0.296
median_hh_income	100,819	95,047	93,849	67,544	+0.291
nss_per_eqv	0.07	0.09	0.12	0.20	+0.282
sped_pct	18.8	16.4	21.9	21.9	+0.273

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	2.9	4.1	-1.2pp
Marlborough	1.4	4.5	+3.1pp better
Clinton	1.1	4.2	+3.1pp better
Worcester	2.1	5.0	+2.9pp better
Palmer	0.9	3.4	+2.5pp better
Norwood	0.1	2.5	+2.4pp better
Fitchburg	2.1	4.5	+2.4pp better

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Dropout Rate

Predicted 4.1pp · Actual 2.9pp · Gap -1.2pp (lower is better) · Saugus ranks 27 of 168 MA districts on this measure

The Bottom Line

Saugus's dropout rate is 2.9%, 1.2pp better than its demographic prediction of 4.1% — ranking 27 of 168 MA districts, in the top 25%. Dropout reflects re-engagement and attendance recovery as much as demographics.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.91
Teachers per 100 students	5.87	8.07	+0.39
Adults with a bachelor's degree+	31.4%	46.9%	+0.38
Teachers per 100 ÷ low-income %	0.126	0.277	+0.33
Health insurance \$ per resident	\$588	\$453	+0.33

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	24.3%	-6.9pp
Teachers per 100 students	5.87	7.46	+1.59
Adults with a bachelor's degree+	31.4%	34.6%	+3.1pp
Teachers per 100 ÷ low-income %	0.126	0.133	+0.007
Health insurance \$ per resident	\$588	\$632	+44

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

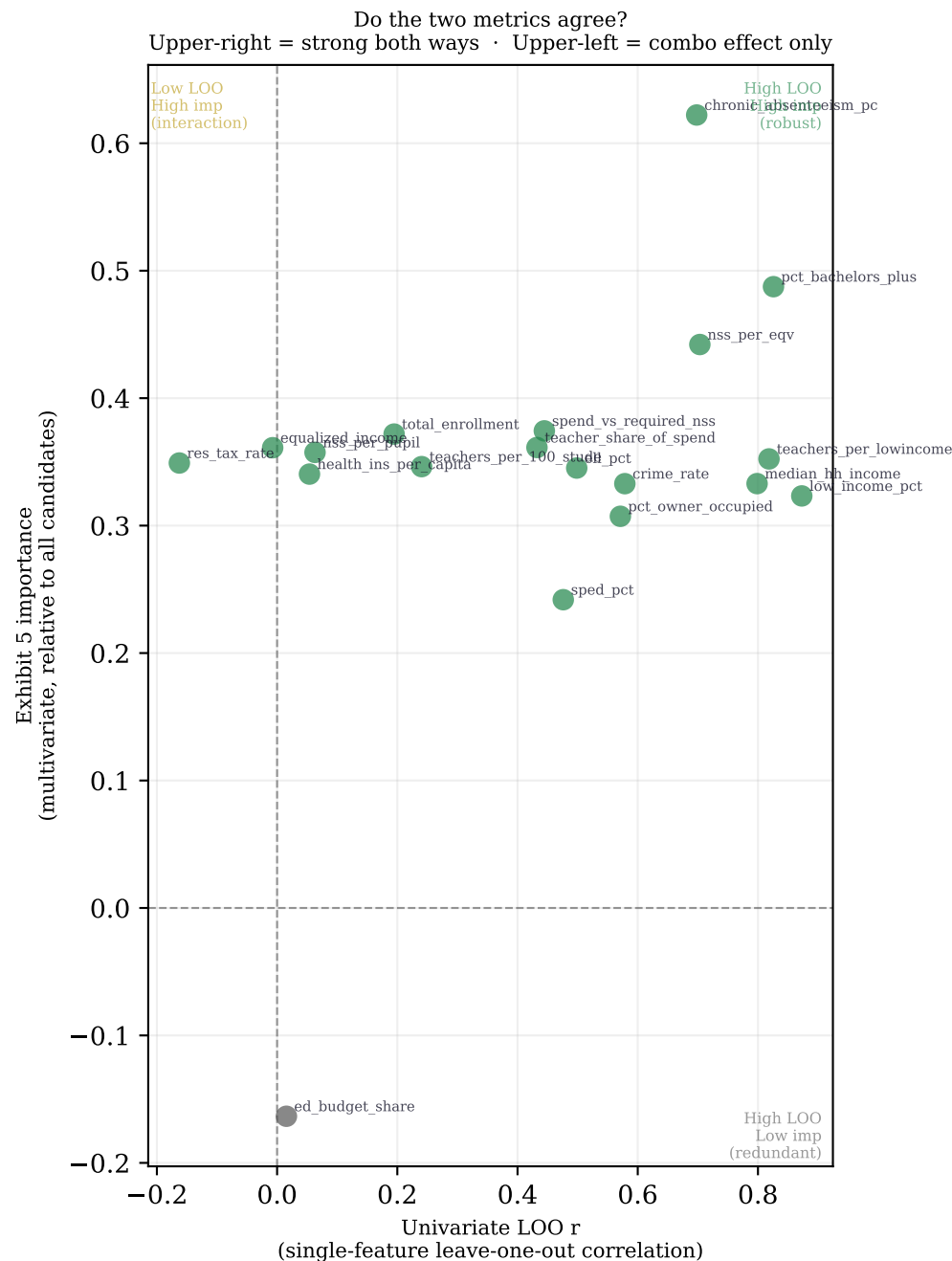
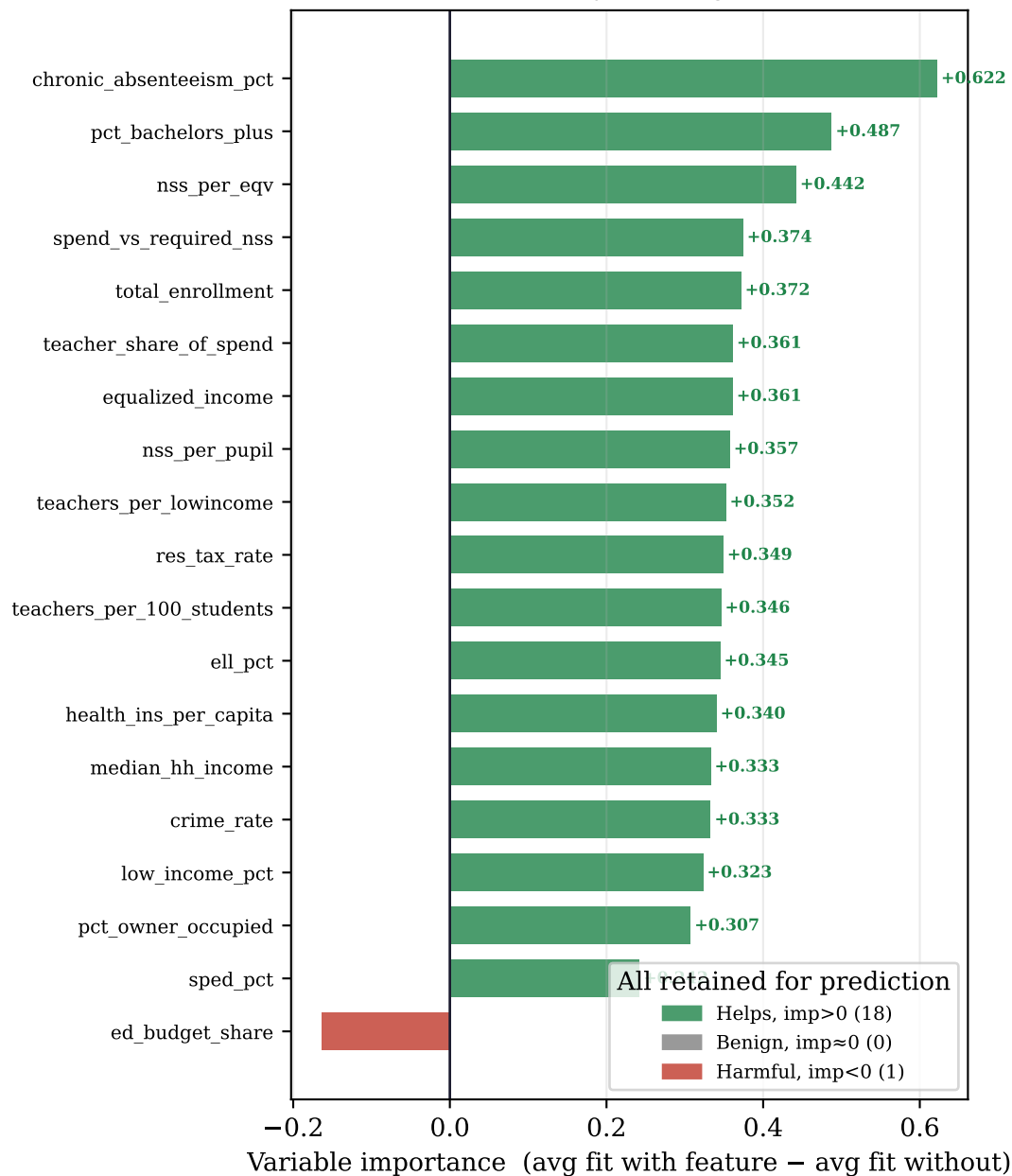
The Takeaway

Saugus does better here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is health insurance \$ per resident (+44, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) Chronic absenteeism is the factor to watch — the main risk to this result if it climbs.

Variable Importance (Exhibit 5): MCAS Grade 10 (ELA)

19 candidates, all used for prediction · n_random=3000 · top-3 NOT seed-stable · 18 help (imp>0) · 0 benign (≈0) · 1 harmful (imp<0)

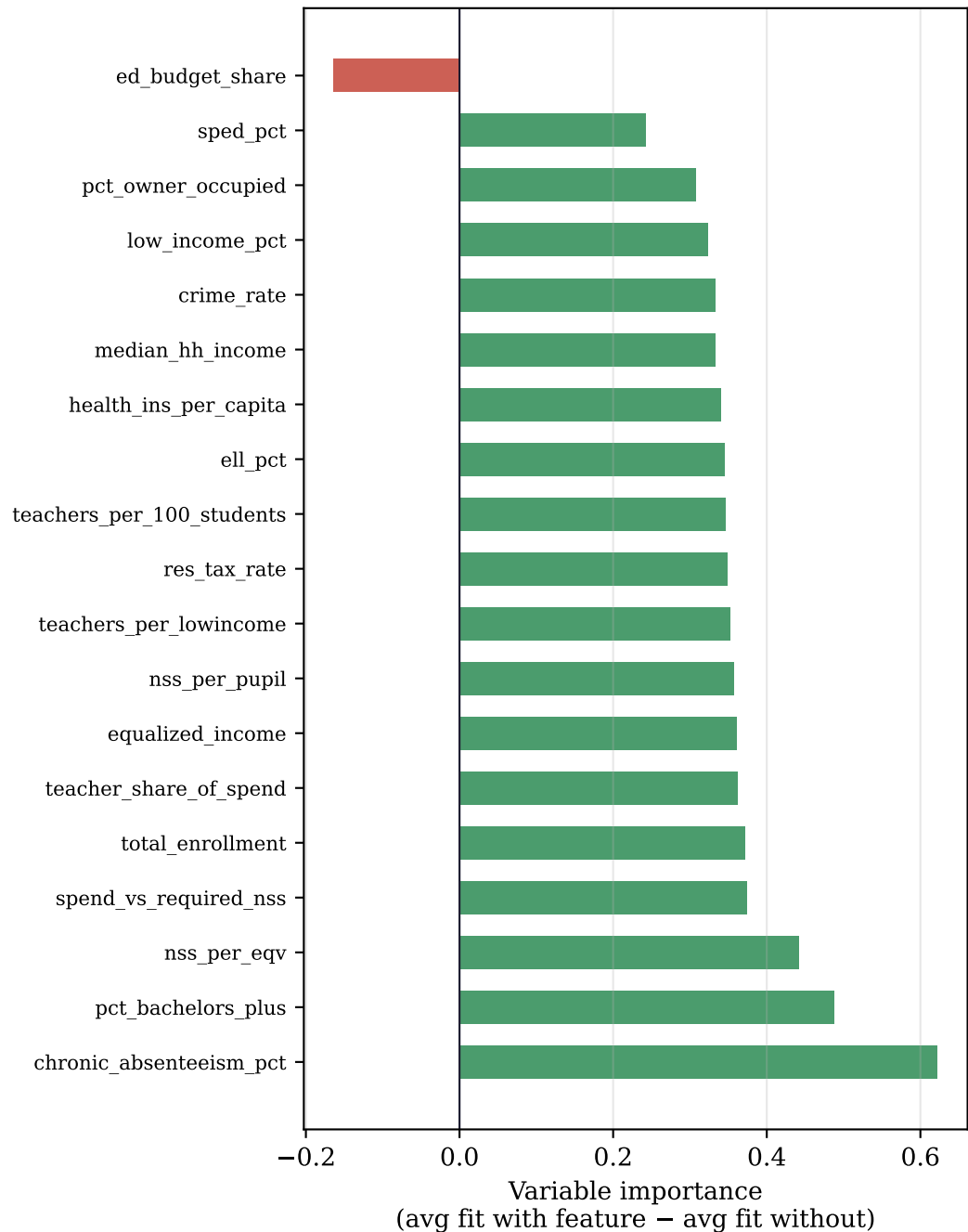
All Candidates — Variable Importance (all retained)
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful



Saugus RBP Analysis: MCAS Grade 10 (ELA)

Prediction task = Saugus · Predicted: 34.9pp Actual: 47.0pp Gap: +12.1pp (Fit: 0.1572)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	mcas10_ela
Everett	0.0837	0.37
Gardner	0.0797	0.40
Winchendon	0.0692	0.33
Gloucester	0.0674	0.38
Dracut	0.0659	0.53

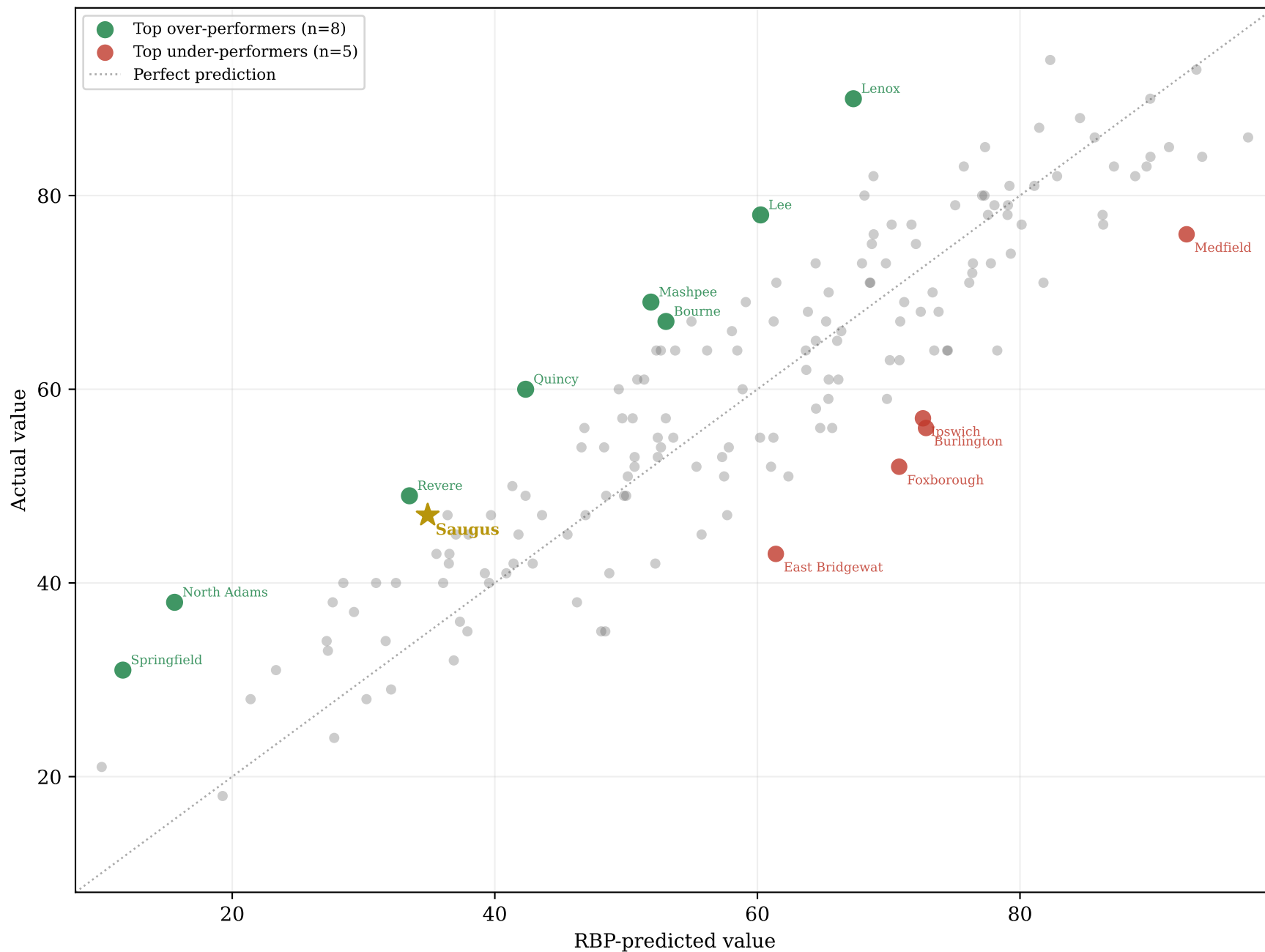
Least Relevant Towns (lowest weight)

Town	Weight	mcas10_ela
Andover	-0.0187	0.79
Bedford	-0.0184	0.83
Foxborough	-0.0184	0.52
Newburyport	-0.0184	0.71
Natick	-0.0174	0.72

Over- and Under-Performers: MCAS Grade 10 (ELA)

Saugus gap: +12.1pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grade 10 (ELA) (LOO r = 0.900)



Residual = actual – predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: MCAS Grade 10 (ELA)

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Lenox	North Ada	Springfie	Imp
chronic_absenteeism_pct	31.2	18.3	30.4	32.4	+0.622
pct_bachelors_plus	31.4	49.5	27.0	19.6	+0.487
nss_per_eqv	0.07	0.06	0.31	0.30	+0.442
spend_vs_required_nss	1.03	1.54	1.45	1.26	+0.374
total_enrollment	3,084	814.0	1,447	29,468	+0.372
teacher_share_of_spend	0.35	0.37	0.30	0.29	+0.361
equalized_income	1.17B	310M	248M	2.89B	+0.361
nss_per_pupil	17,369	23,945	26,299	26,616	+0.357
teachers_per_lowincome	0.13	0.41	0.12	0.09	+0.352
res_tax_rate	10.7	9.07	17.1	16.1	+0.349
teachers_per_100_student	5.87	10.1	8.67	7.37	+0.346
ell_pct	13.0	4.10	0.60	17.1	+0.345
health_ins_per_capita	588.1	890.8	521.6	534.9	+0.340
median_hh_income	100,819	97,569	48,521	51,339	+0.333
crime_rate	2,813	2,695	7,724	8,923	+0.333

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	47.0	34.9	+12.1pp
Lenox	90.0	67.3	+22.7pp
North Adams	38.0	15.6	+22.4pp
Springfield	31.0	11.7	+19.3pp
Lee	78.0	60.2	+17.8pp
Quincy	60.0	42.3	+17.7pp
Mashpee	69.0	51.9	+17.1pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grade 10 (ELA)

Predicted 34.9pp · Actual 47.0pp · Gap +12.1pp (higher is better) · Saugus ranks 10 of 168 MA districts on this measure

The Bottom Line

Saugus's grade 10 ELA proficiency is 47.0%, 12.1pp better than its demographic prediction of 34.9% — ranking 10 of 168 MA districts, in the top 10% of MA districts. Note: the grade 3–8 academic base is an outcome, not a lever, so it is excluded from this model — part of the over-performance shown here reflects that omitted prior strength rather than a grade-10 effect.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.62
Adults with a bachelor's degree+	31.4%	46.9%	+0.49
School spending ÷ property wealth/capita	0.068	0.082	+0.44
Spending vs Ch70 required minimum (×)	1.03	1.36	+0.37
Total district enrollment	3,084	1,669	+0.37

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	18.9%	-12.3pp
Adults with a bachelor's degree+	31.4%	44.7%	+13.2pp
School spending ÷ property wealth/cap	0.068	0.090	+0.023
Spending vs Ch70 required minimum (×)	1.03	1.40	+0.369
Total district enrollment	3,084	1,705	-1,379

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

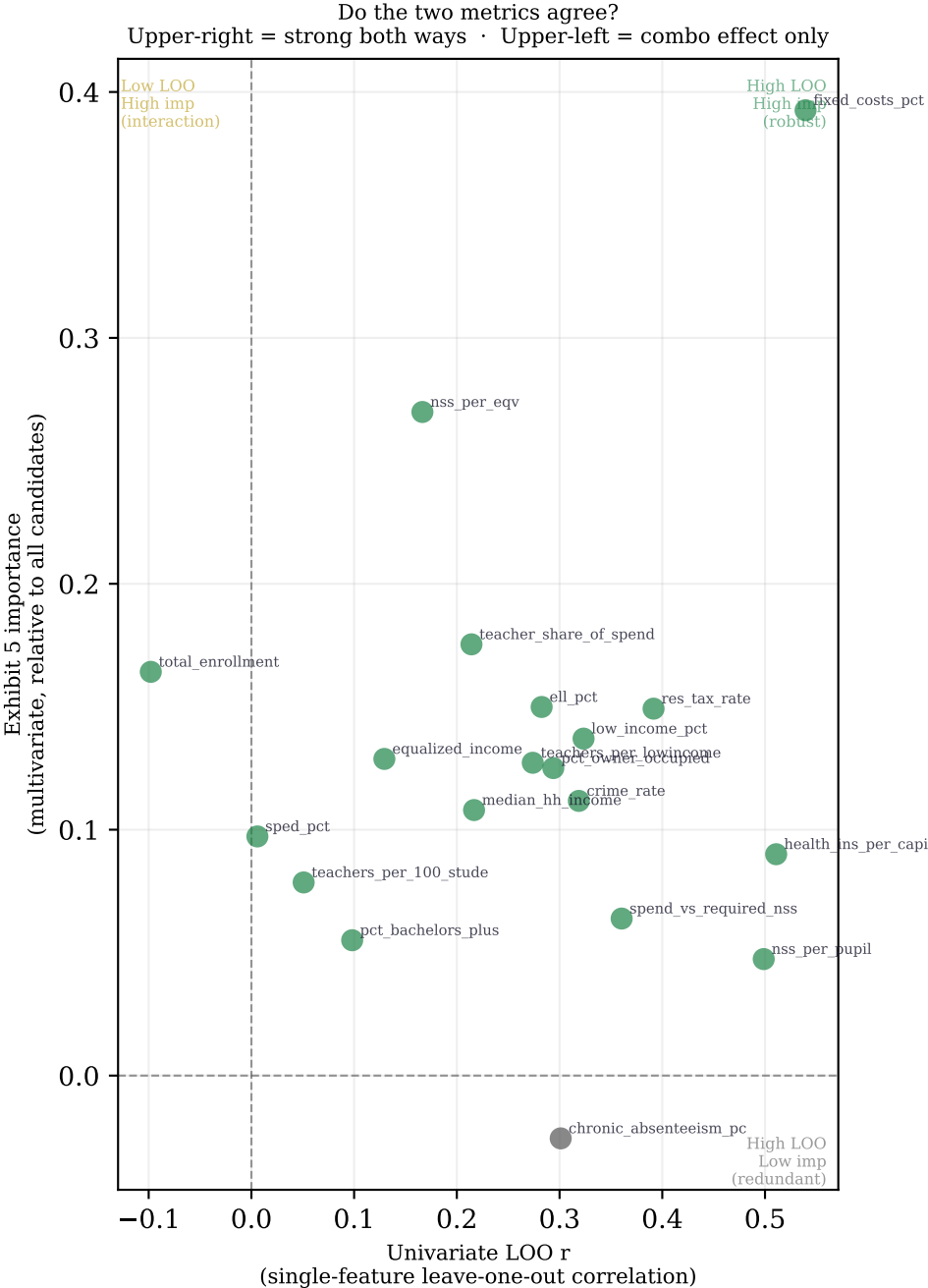
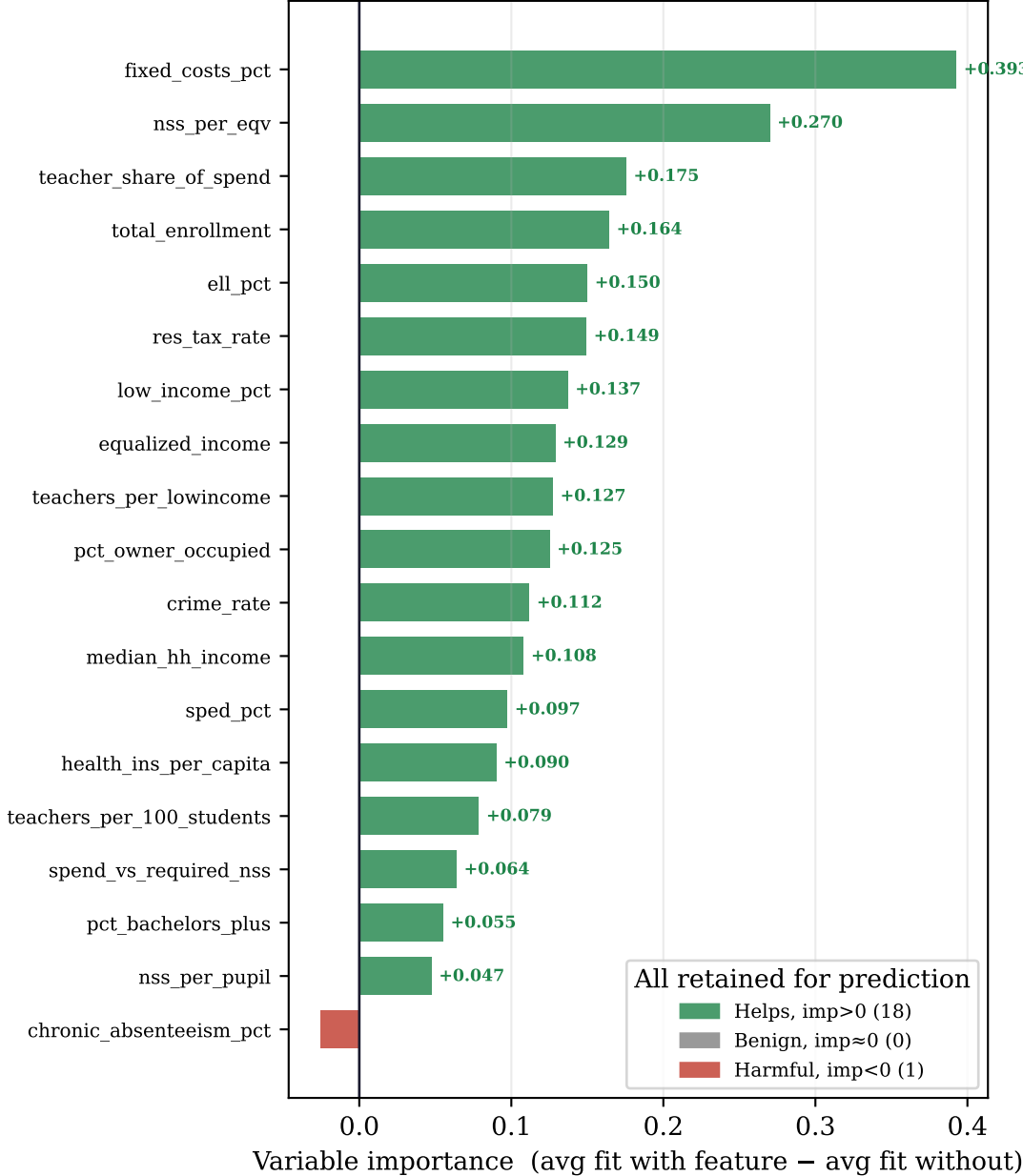
The Takeaway

Saugus does better here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is total district enrollment (-1,379, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) Among the actionable drivers, chronic absenteeism and teacher staffing are the most visible — Saugus clears this bar despite lagging peers on both.

Variable Importance (Exhibit 5): Education Budget Share

19 candidates, all used for prediction · n_random=3000 · top-3 NOT seed-stable · 18 help (imp>0) · 0 benign (≈0) · 1 harmful (imp<0)

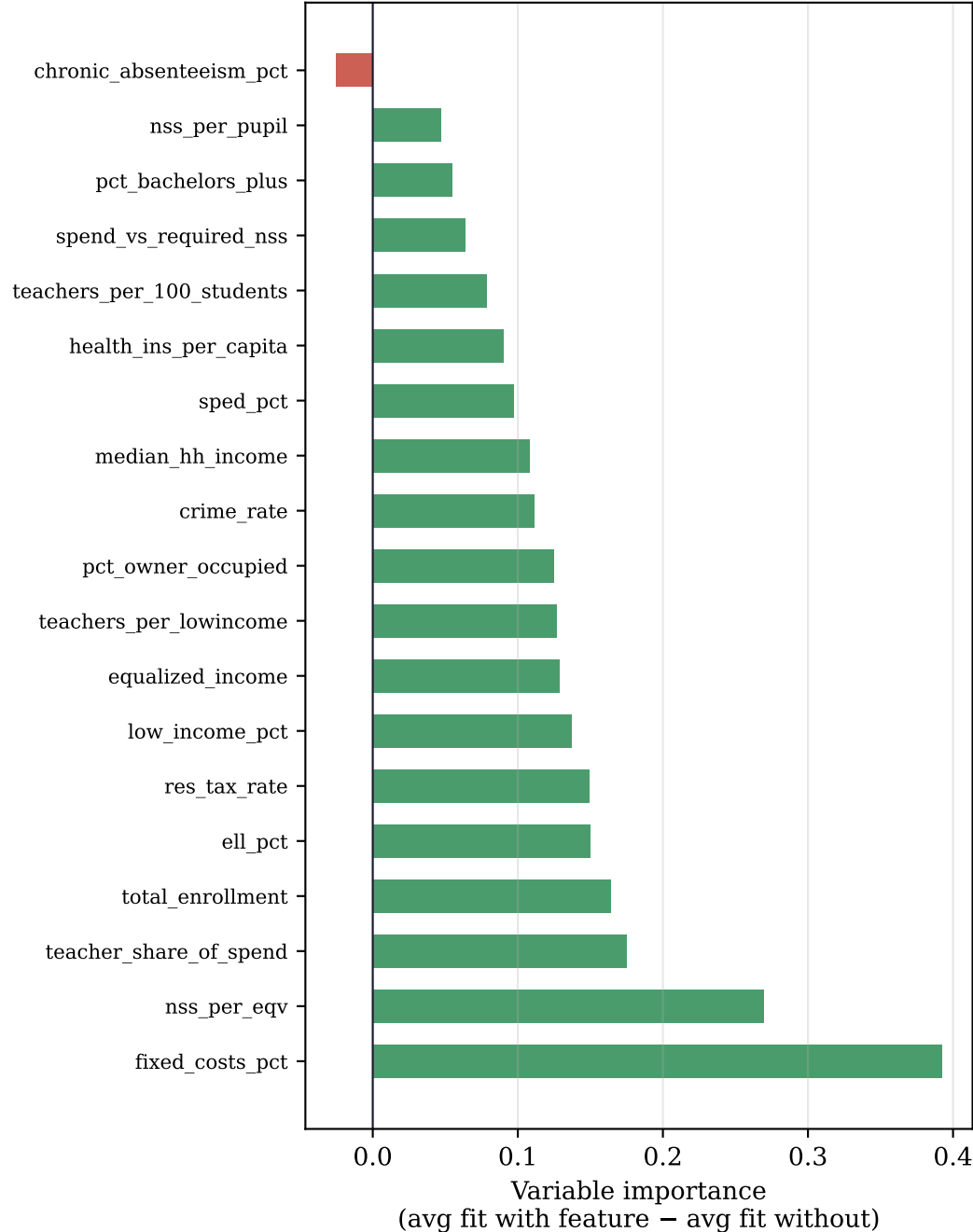
All Candidates — Variable Importance (all retained)
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful



Saugus RBP Analysis: Education Budget Share

Prediction task = Saugus · Predicted: 40.9pp Actual: 32.3pp Gap: -8.6pp (Fit: 0.0844)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

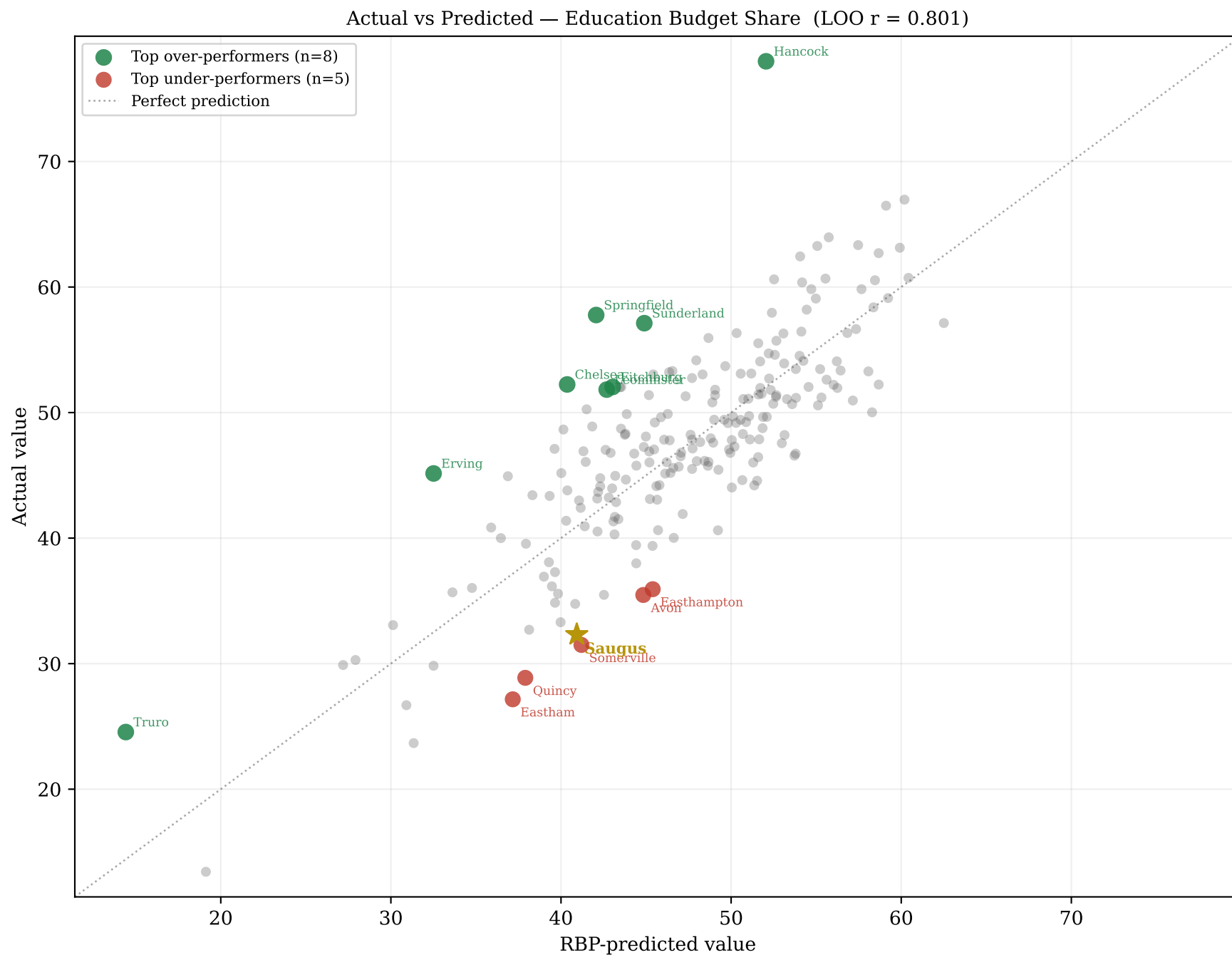
Town	Weight	ed_budget_sh
Everett	0.0456	41.38
Webster	0.0412	50.26
Dracut	0.0386	47.85
Nahant	0.0385	32.70
Gardner	0.0371	43.68

Least Relevant Towns (lowest weight)

Town	Weight	ed_budget_sh
Amherst	-0.0176	52.21
Shutesbury	-0.0126	60.61
Sunderland	-0.0122	57.12
Dedham	-0.0116	40.62
Bedford	-0.0115	44.62

Over- and Under-Performers: Education Budget Share

Saugus gap: -8.6pp · Gold star = Saugus · Green = over-performers · Red = under-performers



Residual = actual – predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: Education Budget Share

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Hancock	Springfie	Erving	Imp
fixed_costs_pct	25.3	2.43	19.7	16.3	+0.393
nss_per_eqv	0.07	0.04	0.30	0.06	+0.270
teacher_share_of_spend	0.35	0.38	0.29	0.29	+0.175
total_enrollment	3,084	106.0	29,468	208.0	+0.164
ell_pct	13.0	0.00	17.1	0.80	+0.150
res_tax_rate	10.7	2.79	16.1	9.22	+0.149
low_income_pct	46.6	36.2	85.0	44.7	+0.137
equalized_income	1.17B	13M	2.89B	46M	+0.129
teachers_per_lowincome	0.13	0.20	0.09	0.21	+0.127
pct_owner_occupied	73.9	77.8	49.5	76.6	+0.125
crime_rate	2,813	0.00	8,923	3,020	+0.112
median_hh_income	100,819	93,125	51,339	74,550	+0.108
sped_pct	18.8	24.1	26.8	21.4	+0.097
health_ins_per_capita	588.1	11.3	534.9	894.5	+0.090
teachers_per_100_student	5.87	7.36	7.37	9.33	+0.079

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	32.3	40.9	-8.6pp
Hancock	78.0	52.1	+25.9pp
Springfield	57.8	42.1	+15.7pp
Erving	45.1	32.5	+12.6pp
Sunderland	57.1	44.9	+12.2pp
Chelsea	52.2	40.4	+11.9pp
Truro	24.6	14.4	+10.1pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Education Budget Share

Predicted 40.9pp · Actual 32.3pp · Gap -8.6pp (higher is better) · Saugus ranks 214 of 219 MA districts on this measure

The Bottom Line

Saugus's education budget share is 32.3%, 8.6pp worse than its demographic prediction of 40.9% — ranking 214 of 219 MA districts, in the bottom 10%. Unlike the academic measures, this gap reflects a municipal budgeting choice, not a demographic constraint.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Pension & benefits share of budget	25.3%	17.5%	+0.39
School spending ÷ property wealth/capita	0.068	0.082	+0.27
Teacher pay share of school spending	0.349	0.368	+0.18
Total district enrollment	3,084	1,669	+0.16
English language learners	13.0%	3.4%	+0.15

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Pension & benefits share of budget	25.3%	16.1%	-9.2pp
School spending ÷ property wealth/cap	0.068	0.137	+0.069
Teacher pay share of school spending	0.349	0.315	-0.035
Total district enrollment	3,084	3,052	-32
English language learners	13.0%	11.3%	-1.7pp

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

Saugus does worse here than its demographic profile predicts. The strongest RBP driver of Saugus's predicted peer group is pension & benefits share of budget (25.3% vs the 17.5% state median); the widest gap to the over-performing districts is total district enrollment (-32, peer median – Saugus). (Driver ranking is close enough at the top that the #1 factor is not fully seed-stable — read the top group, not the exact order.) The top drivers are competing budget categories that crowd out the education share — the lever here sits with Town Meeting and municipal budgeting, not the school district.

Combined Model Summary: All Four Outcomes

RBP factor selection results across MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, and Education Budget Share

Model	Predicted	Actual	Gap	Direction	LOO r	Features
MCAS Grades 3–8	23.6%	29.0%	+5.4pp	over	+0.902	19
Dropout Rate	4.1%	2.9%	-1.2pp	over	+0.838	19
MCAS Grade 10 (ELA)	34.9%	47.0%	+12.1pp	over	+0.900	19
Education Budget Share	40.9%	32.3%	-8.6pp	under	+0.801	19

Feature Cross-Reference — Which Factors Earn Their Place Across Models

A feature appearing in multiple models is a robust signal, not a model-specific artifact.

Feature	# Models	MCAS 3–8	Dropout	MCAS 10	Ed Budget
chronic_absenteeism_pct	4	+0.541	+0.908	+0.622	-0.026
pct_bachelors_plus	4	+0.506	+0.376	+0.487	+0.055
nss_per_eqv	4	+0.306	+0.282	+0.442	+0.270
total_enrollment	4	+0.400	+0.332	+0.372	+0.164
teacher_share_of_spend	4	+0.378	+0.327	+0.361	+0.175
equalized_income	4	+0.391	+0.333	+0.361	+0.129
teachers_per_lowincome	4	+0.375	+0.335	+0.352	+0.127
teachers_per_100_students	4	+0.359	+0.387	+0.346	+0.079
ell_pct	4	+0.359	+0.296	+0.345	+0.150
res_tax_rate	4	+0.322	+0.325	+0.349	+0.149
low_income_pct	4	+0.416	+0.267	+0.323	+0.137
spend_vs_required_nss	4	+0.377	+0.314	+0.374	+0.064
crime_rate	4	+0.332	+0.324	+0.333	+0.112
median_hh_income	4	+0.341	+0.291	+0.333	+0.108
health_ins_per_capita	4	+0.300	+0.335	+0.340	+0.090
pct_owner_occupied	4	+0.287	+0.270	+0.307	+0.125
nss_per_pupil	4	+0.326	+0.233	+0.357	+0.047
sped_pct	4	+0.195	+0.273	+0.242	+0.097
ed_budget_share	3	-0.116	-0.218	-0.163	—
fixed_costs_pct	1	—	—	—	+0.393

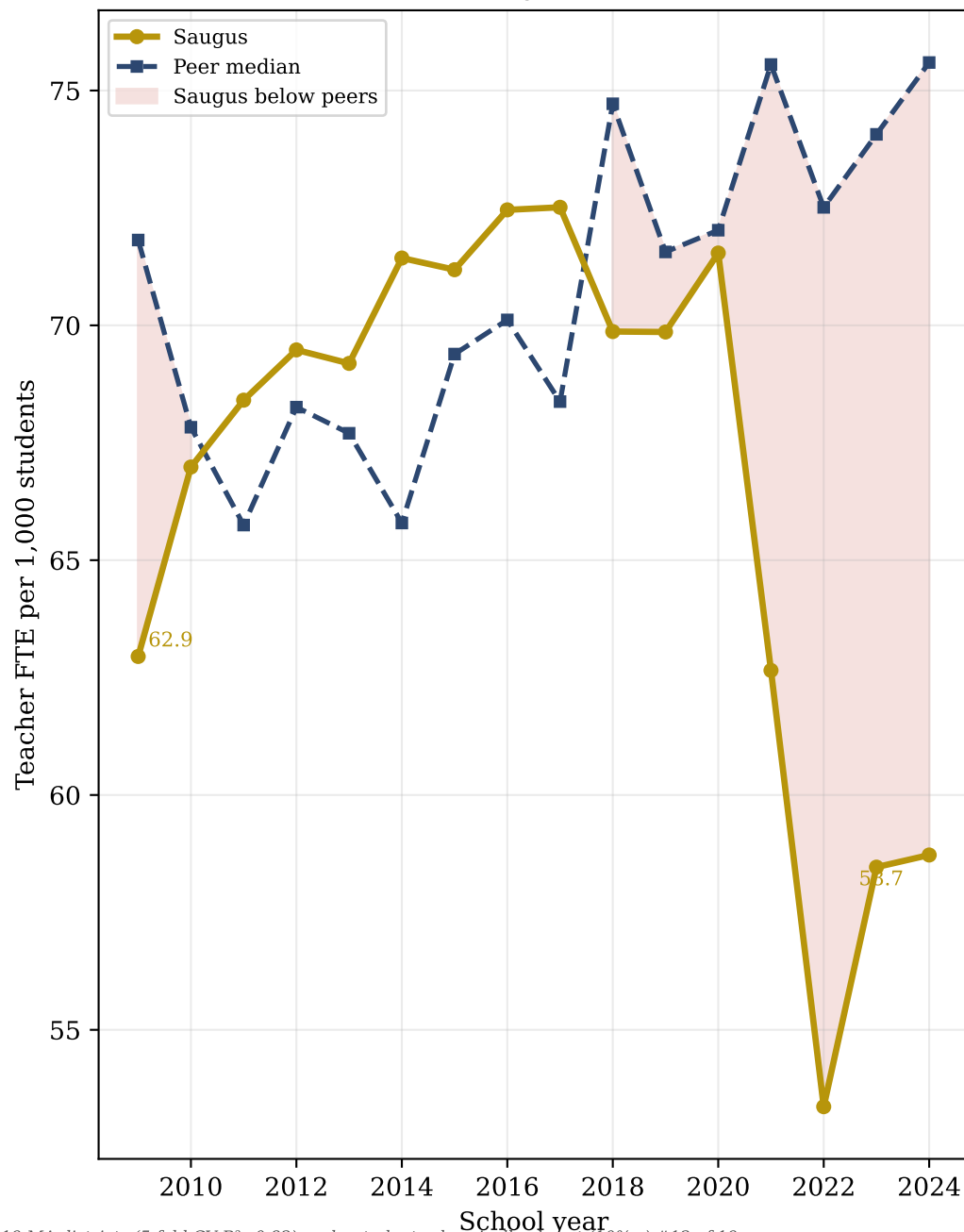
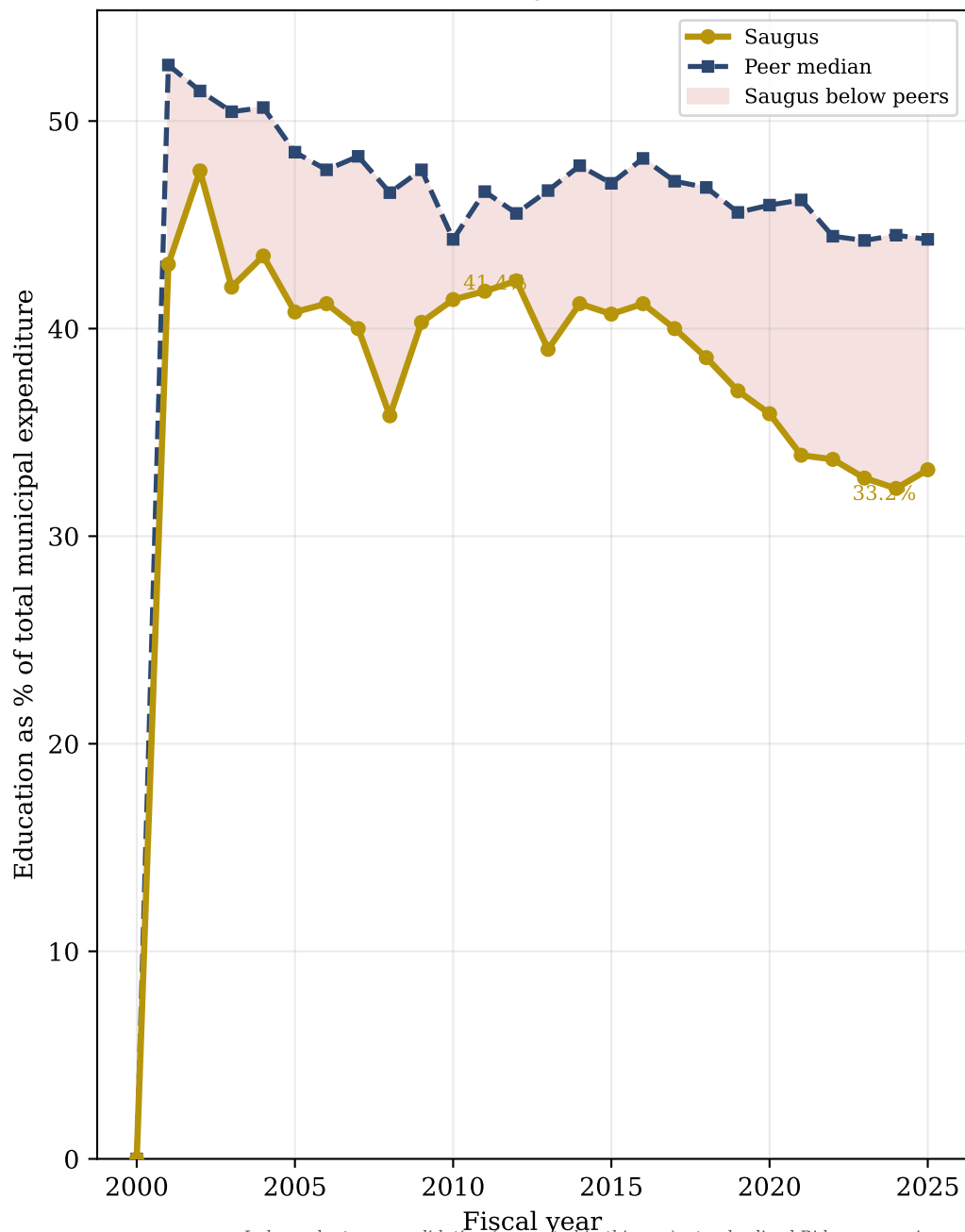
Green rows = feature appears in 3+ models. Cell colour: green = positive importance, red = negative importance, white = not in that model (—).

Budget Allocation & Teacher Density — 15-Year Trajectory

Source: MA DLS Schedule A (budget) · MA DESE Staffing (teachers) · Peer comparison: 8 demographically similar overachiever towns

Education Budget Share
(Schedule A, general fund)

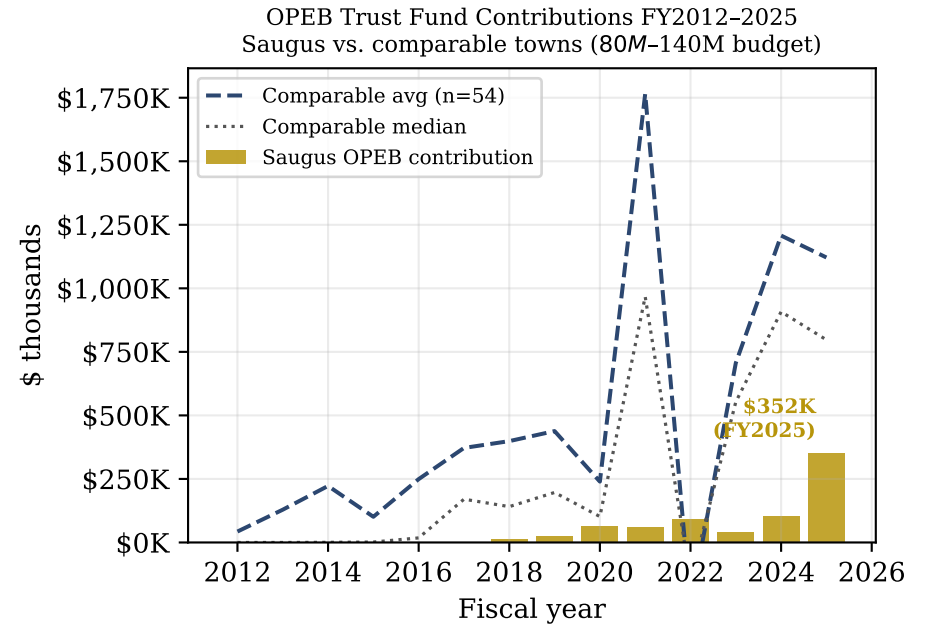
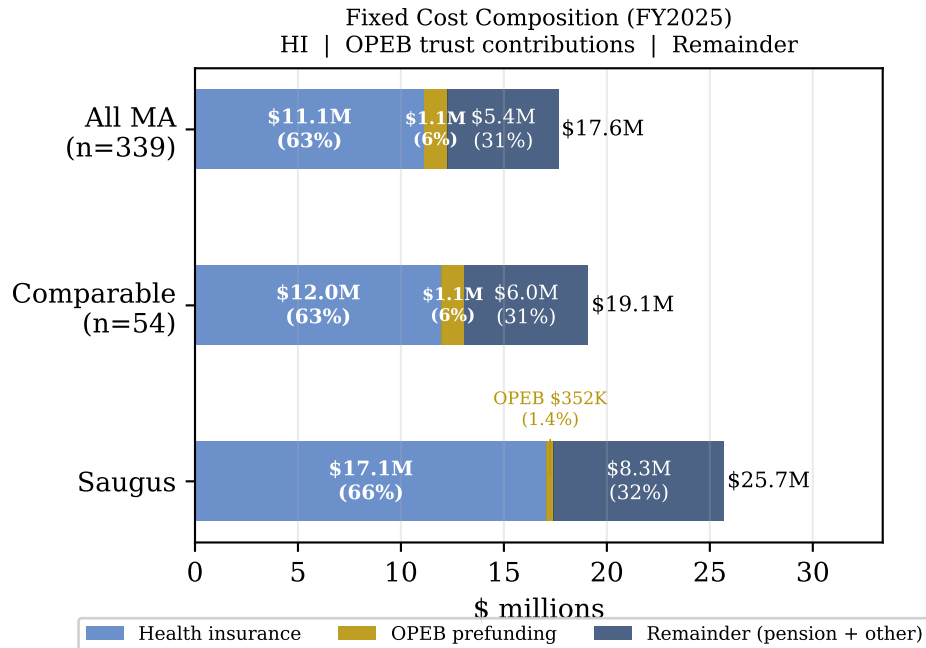
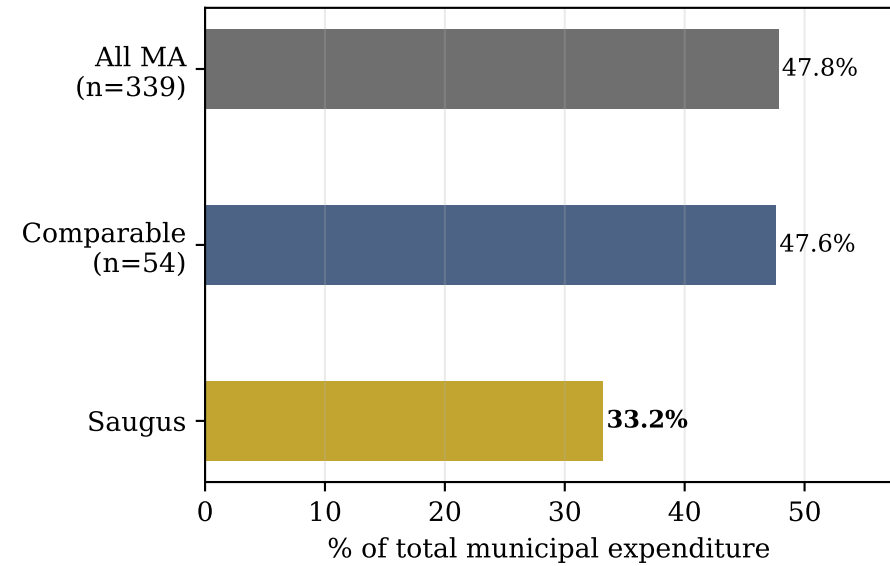
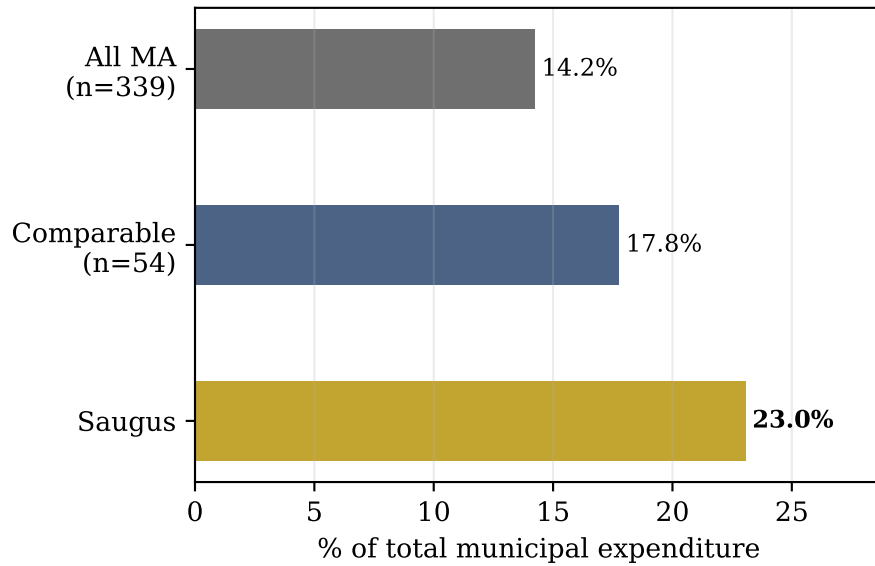
Teacher Density
(DESE Staffing + Enrollment)



Independent cross-validation (computed in this run): standardized Ridge regression on 219 MA districts (5-fold CV $R^2=0.82$) ranks students chronically absent (10%+) #12 of 19 standardized predictors of MCAS 3–8 ($|standardized\ \beta|=0.08$) — a divergence, not corroboration: a linear model assigns students chronically absent (10%+) a small standardized coefficient, whereas RBP's Exhibit 5 ranks it among the top drivers — the kind of nonlinear / interaction effect RBP is designed to surface and linear regression misses. Peer set: 8 demographically similar overachiever towns (Mahalanobis $d_M \leq 2$).

Fixed Costs: Sources, Scale, and OPEB Prefunding

FY2025 · Comparable = 54 MA towns (80M-140M) · All MA = 339 · Columns sum exactly to total for all 4,898 statewide municipality-years



Optimum Profile — What Overachievers Look Like

Each grid shows actionable feature values for Saugus vs overachiever towns. Showing both peer groups tests whether the finding holds across definitions.

Overachievers — all 20 MA districts that beat their demographic prediction

Feature	Saugus	Springfie	North Ada	Fitchburg	Quincy	Amherst	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	32.4%	30.4%	34.0%	16.5%	16.7%	19.9%	-11.3pp	20
Teachers / 100 students	5.9	7.4	8.7	7.6	7.5	9.8	8.3	+2.4	20
Avg teacher salary	\$89,778	\$84,692	\$76,435	\$84,723	\$94,955	\$87,842	\$84,708	\$-5,070	20
Net school spending / pupil	\$17,369	\$26,616	\$26,299	\$20,866	\$20,257	\$31,268	\$22,700	+\$+5,330	20
Shown: Springfield, North Adams, Fitchburg, Quincy, Amherst (+ 15 more in median)									

Demographically Similar Overachievers — 17 towns within 2σ Mahalanobis distance of Saugus on income, property wealth, poverty, size, education & ELL

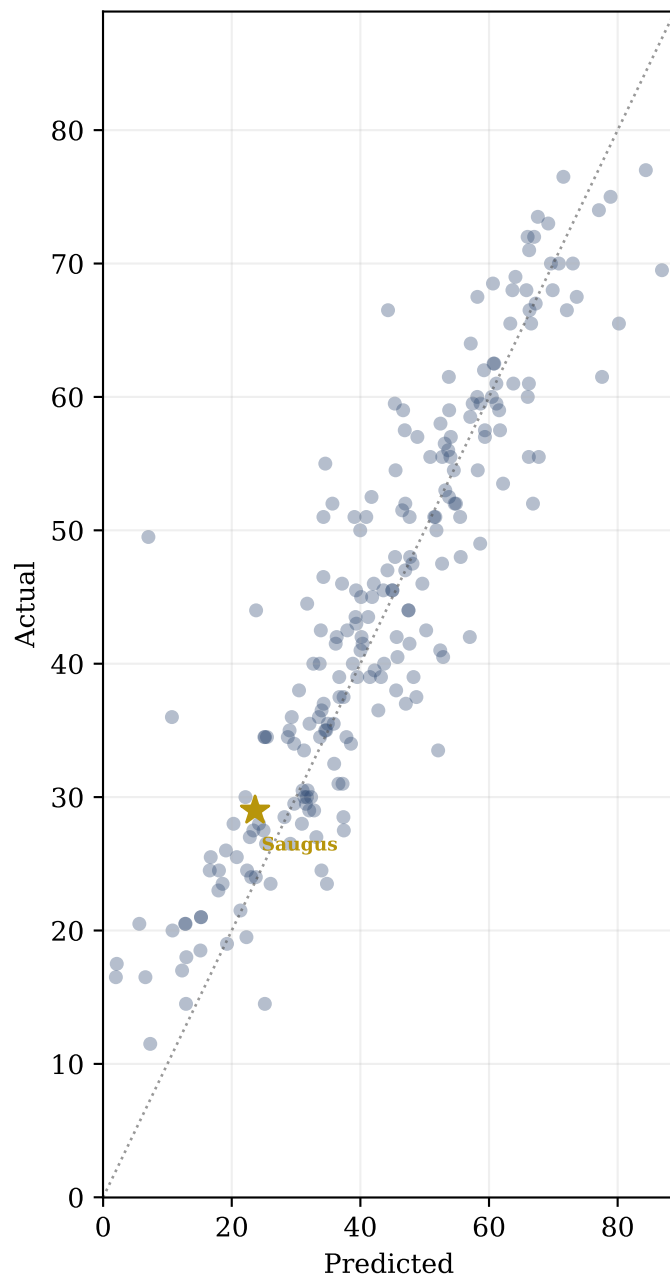
Feature	Saugus	Norwood	Clinton	Westport	Lenox	Acushnet	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	25.1%	7.7%	13.8%	18.3%	9.2%	18.3%	-12.9pp	17
Teachers / 100 students	5.9	8.8	7.3	7.5	10.1	6.1	8.4	+2.5	17
Avg teacher salary	\$89,778	\$92,324	\$87,059	\$83,017	\$84,669	\$77,654	\$84,723	\$-5,055	17
Net school spending / pupil	\$17,369	\$21,084	\$18,512	\$18,099	\$23,945	\$16,414	\$21,084	+\$+3,715	17
Shown: Norwood, Clinton, Westport, Lenox, Acushnet (+ 12 more in median)									

Similarity metric: Mahalanobis distance $d_M = \sqrt{(\Delta x \cdot \Sigma^{-1} \cdot \Delta x')}$, Σ = sample covariance of all MA districts on (median_hh_income, equalized_income, low_income_pct, total_enrollment, pct_bachelors_plus, ell_pct). Threshold: $d_M \leq 2$ (2σ). Pre-specified; not tuned to result. Gap = peer median – Saugus. Red = Saugus below peer median, Green = at or above.

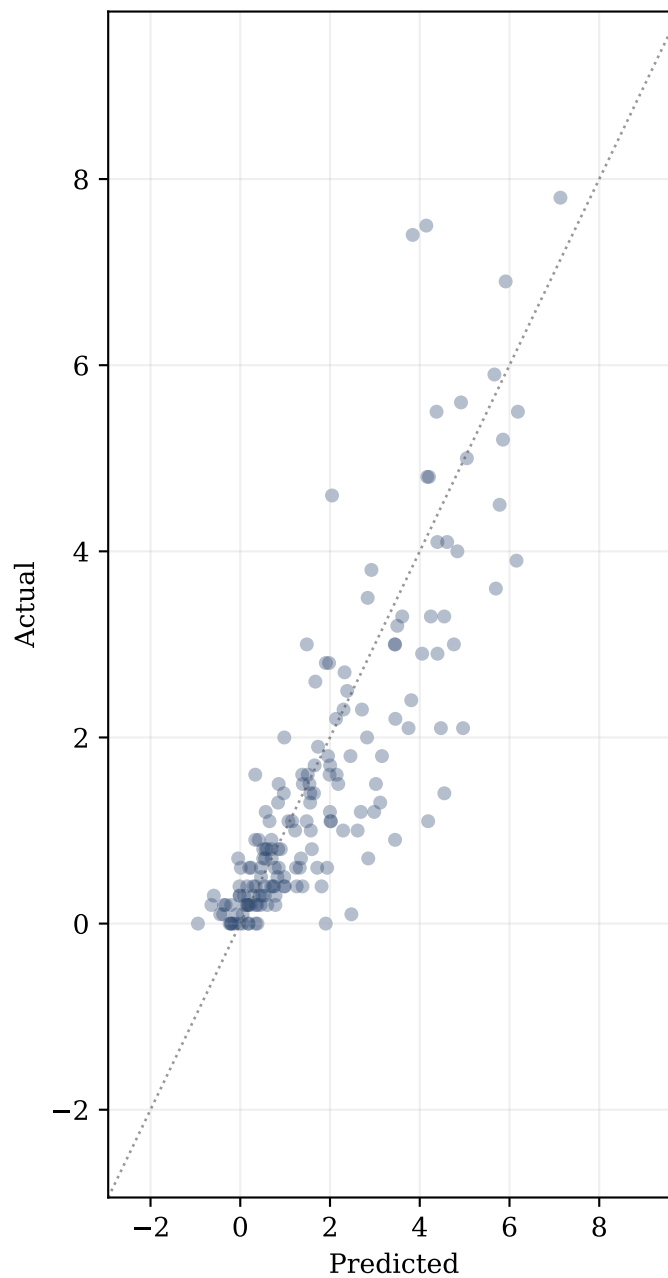
Actual vs. Predicted: All Three Models

Each dot = one MA district. Saugus highlighted in gold.

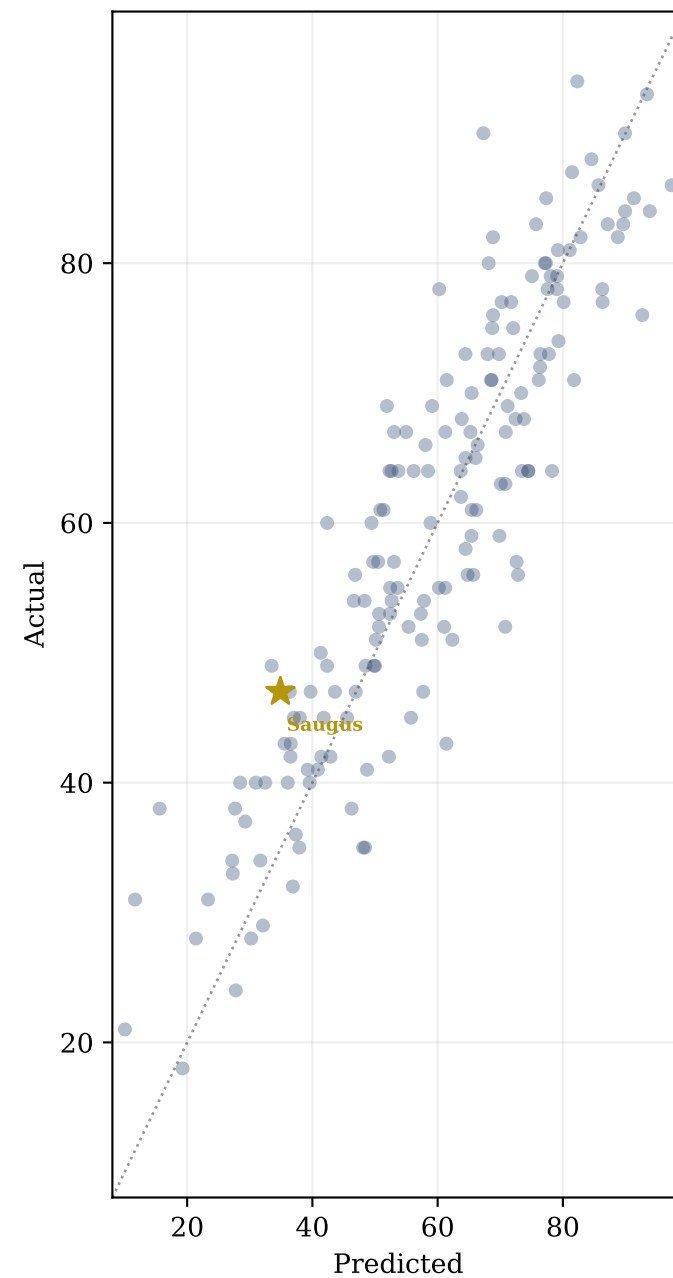
MCAS Grades 3-8
LOO r = 0.902



Dropout Rate
LOO r = 0.838



MCAS Grade 10 (ELA)
LOO r = 0.900



Leave-one-out predictions — each district excluded from its own prediction.